

Publishing Linked Open Data from Spreadsheets

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Abstract

The World Wide Web (WWW), as defined by the World Wide Web Consortium (W3C), is “the universe of network-accessible information, the embodiment of human knowledge”¹. With development over the years, the global web of linked documents has become the global web of linked data, which is popularly known as the Semantic Web.

The Semantic Web is now an established topic of research but there remain many unanswered questions. Data that are linked and open facilitate discovery and reuse. Likewise, vocabularies used to describe the semantic relationships amongst linked and open data facilitate discovery and reuse. Yet, it is not always a straightforward matter to take data that may be somehow accessible on a webpage written in Hyper Text Markup Language (HTML) and publish it as Linked Open Data (LOD). Many complementary technologies are used in handling LOD on the web, particularly the Resource Description Framework (RDF), the Web Ontology Language (OWL), and the Simple Protocol and RDF Query Language (SPARQL), to name a few.

Tim Berners-Lee’s vision for a global hypertext system continues to be fulfilled. This thesis work contributes to the practice of publishing LOD by following the lifecycle of a particular dataset from spreadsheet to LOD that is accessible on the web. As such, this thesis can serve as a resource for researchers and citizens-at-large who want to contribute their data to the world.

¹<https://www.w3.org/WWW/>

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List of Abbreviations

CERN Conseil Européen pour la Recherche Nucléaire

DL Description Logics

ER Entity Relationship

GREL Google Refine Expression Language

HTML Hyper Text Markup Language

HTTP Hyper Text Transfer Protocol

HTTPS Hyper Text Transfer Protocol Secure

IRI Internationalized Resource Identifier

ISO International Organization for Standardization

LOD Linked Open Data

N3 Notation3

OWL Web Ontology Language

RDF Resource Description Framework

RDFS Resource Description Framework Schema

RML RDF Mapping Language

SHACL SHApes Constraint Language

SPARQL Simple Protocol and RDF Query Language

SWT Semantic Web Technologies

Turtle Terse RDF Triple Language

URI Uniform Resource Identifier

W3C World Wide Web Consortium

WWW World Wide Web

XML eXtensible Markup Language

Chapter 1

Introduction

Tim Berners-Lee is the father of the WWW. He wrote *Information Management: A Proposal*¹ in 1989 and 1990 to present the need for a “global hypertext system” at the Conseil Européen pour la Recherche Nucléaire (CERN). In his conclusion to that document, he wrote:

“we should work toward a universal linked information system, in which generality and portability are more important than fancy graphics techniques and complex extra facilities. The aim would be to allow a place to be found for any information or reference which one felt was important and a way of finding it afterward. The result should be sufficiently attractive to use, that is, the information contained would grow past a critical threshold so that the usefulness of the scheme would, in turn, encourage its increased use.”

In the more than 30 years that have followed his initial proposal, a great deal of progress towards Berners-Lee’s vision has been realized. The original “web of documents” is becoming the “web of data” or the “semantic web”.

The gold standard for data on the semantic web is 5-star LOD. All resources (subjects, predicates, and objects) are named with Hyper Text Transfer Protocol Secure (HTTPS) Uniform Resource Identifiers (URIs) that can be looked-up on the web to

¹<https://www.w3.org/History/1989/proposal.html>, dated March 1989, May 1990

provide useful information with the RDF family of standards. This data is accompanied with an open license for access.

Is this next web of data sufficiently attractive to use? This thesis will examine this question in the context of an example.

1.1 Motivation

In his teaching practise, my supervisor collects feedback from students in his classes each semester and makes the responses available on his website². Typically, student responses are collected on paper and then entered manually into computer-readable form. Data collection was changed and interrupted during the COVID-19 pandemic.

Although it is valuable to have this feedback data available on the web in any form, it is not published as LOD. The motivation for the research presented here is to examine the process by which existing data can be published as LOD and to consider whether that process is “sufficiently attractive to use”.

1.2 Contribution

Within the LOD community, most research effort deals with users who are domain experts. This thesis is focused on the usability of tools and frameworks for the development, description and reuse of LOD by citizens.

The research presented here is a practical exploration of the process to take spreadsheet data and publish it on the web as 5-star LOD. More than converting the data into RDF, it is about conveying as much as possible of the semantics of the data. This thesis examines the linked data lifecycle from the perspective of usability for citizens. Hepting’s feedback instrument has 9 Likert items with a 5-point Likert scale [30] and 3

²<http://www2.cs.uregina.ca/~hepting/teaching/evaluation.html>

open-ended questions. What is the best way to express the semantics in the feedback response data? After a thorough search for Likert scales and items, no vocabularies were found for reuse. A preliminary vocabulary to capture these semantics has been developed and used to represent the feedback response data. The publishing process has been aided by open-source tools, namely Protégé [6] and OpenRefine [18]. Illumination of the practical solutions employed to address problems that arise while using these tools also represents an important contribution.

1.3 Organization

Following this introduction, the rest of the thesis is organized as follows. Chapter 2 describes related work that is pertinent to the thesis. In particular, concepts of the semantic web, linked open data, and related standards and technologies are presented. Chapter 3 describes the linked data lifecycle and some of the tools required. Chapter 4 details the experience of applying the linked data lifecycle to real data. Finally, Chapter 5 presents some conclusions and opportunities for future work.

Chapter 2

Related Work

This Chapter reviews the theoretical and practical work on the semantic web and LOD, from the perspectives of technology and usability, that form the foundation of this thesis.

The web was first designed by Berners-Lee in 1989¹ while working at the CERN. On the original web, of documents, the web pages were written and the links between pages were described using HTML. On the next web, of data, the web pages are written using HTML and the relationships amongst entities on those pages are described using RDF. These differences are illustrated in Figure 2.1.

2.1 Linked Open Data

Linked Open Data is the amalgamation of linked data, that expresses the relationships between data, and open data, that grants broad access rights to the data. The relationships amongst data are equally important to the working of the semantic web as the access to the data itself [9].

In 2006, Berners-Lee [8] described four basic rules for publishing and connecting structured datasets across the web. These rules have become known as the linked data

¹<https://www.w3.org/History/1989/proposal.html>

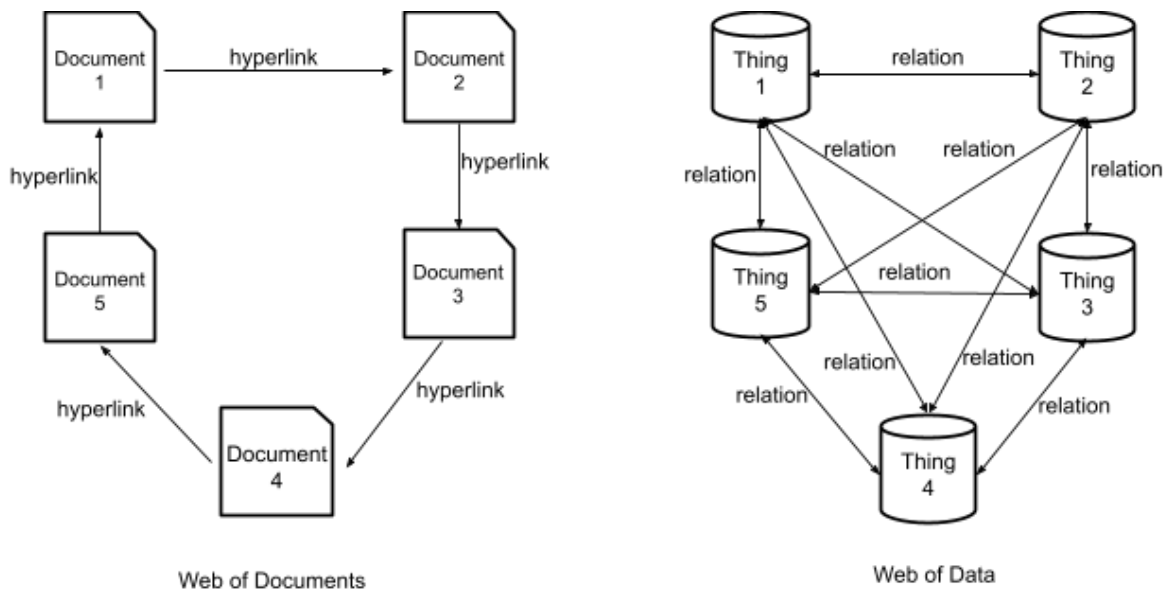


Figure 2.1: Following Aghaei et al. [1], an illustration of the contrast between the web of documents that uses HTML and the web of data that uses RDF.

principles, and have been enhanced by Berners-Lee, Bizer, Heath, and Idehen [7, 9, 10, 26]. Those rules are presented here. Terminology introduced in the rules is described in the subsequent text.

Rule 1. Use URIs as unique names for (all) things on the web, so that the same URI used in different datasets is understood to signify the same thing.

Rule 2. Use HTTPS (or Hyper Text Transfer Protocol (HTTP)) URIs to look up those unique names because the hypertext transfer protocols allow easy retrieval of resources, which promotes the publication and use of data on the web.

Rule 3. Provide useful information to others, using standards from the RDF family, in response to a look-up of a unique name associated with one of your URIs.

Rule 4. Use other (existing) URIs in your data to encourage discovery of more relationships, which promotes the global interconnection of data.

These principles encourage the naming of subjects, predicates, and objects with

HTTPS URIs² and providing useful information through the RDF family of standards. Berners-Lee introduced the concept of 5-star LOD³ to denote structured data on the web as RDF wherein all identifiers are HTTPS URIs pointing to other useful data sources, with an open license for access.

2.1.1 Uniform Resource Identifiers

Resources on the web, whether they are abstract (logical) or physical, are identified by a unique string of ASCII characters known as a URI. As described by Berners-Lee et al. [4] in RFC 3986 (which became STD 66), this unique string is subject to guidelines and security considerations while defining the syntax for a generic URI and a process for resolving URI references. Berners-Lee et al. [4] characterized URI by discussing the component terms.

Numerous advantages come from uniformity. Various types of resource identifier can be used in the same context even when the means accessing those resources may vary. Semantic interpretation of standard syntactic rules is the same across various types of resource identifiers. New types of resource identifiers can be introduced without affecting how other identifiers are used. Existing resource identifiers can be reused in a variety of settings, enabling new applications or protocols to benefit from that widely-utilized set of resource identifiers.

A resource is anything that can be identified by a URI. The pages on a website are a recognizable example. A resource is not necessarily available via the web. It may correspond to a physical entity such as a person, an abstract concept such as a relationship, or a numeric quantity.

An identifier distinguishes an object from all other objects within its scope. The

²<https://www.w3.org/TR/uri-clarification/>

³<https://www.w3.org/TR/ld-glossary/#x5-star-linked-open-data>

purpose of an identifier is to distinguish one resource from all others. An identifier is not required to define the identity of what is referenced, though this may be the true for some. A system using URIs is neither required to access the resource identified and for some there is no intention that they be accessed.

Berners-Lee et al. [4] defined the generic syntax of a URI as shown in Figure 2.2. A URI comprises five components in sequence (see Statements 2.1 and 2.2). It is not mandatory for a URI to have all the components.

$$\text{URI} = \text{scheme:} \quad [//\text{authority}] \text{ path} \quad [?\text{query}] \quad [\#\text{fragment}] \quad (2.1)$$

The authority component comprises three elements, which are:

$$\text{authority} = [\text{userinfo@}] \text{ host} \quad [:\text{port}] \quad (2.2)$$

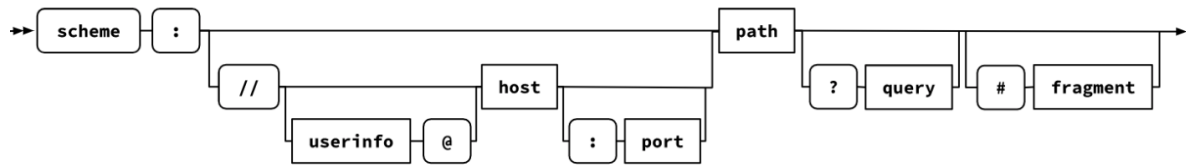


Figure 2.2: Syntax diagram for a generic URI, after Berners-Lee et al. [4].

One of the URIs from the Survey ontology is in Statement 2.3. Figure 2.3 shows its components.

$$\text{https://parita-amin.github.io/survey.owl\#Survey} \quad (2.3)$$



Figure 2.3: Components of the example URI, after Berners-Lee et al. [4]

2.1.2 Resource Description Framework

According to Flanders and Jannidis [19], data modelling refers “the process of representing data, their relationships, and their semantics in a way which is processable by humans and machines”. A data model is an abstract representation of data that standardizes how its components relate to one another and to the characteristics of real-world objects. Depending on the level of abstraction, there are three main types of data models, namely, conceptual, logical, and physical data model. Some widely-used primary data models are dimensional, relational, and Entity Relationship (ER) models. RDF began as a metadata model and is now the means to describe resources (entities) and the relationships amongst them in a computer-understandable format, using basic data models like the ER model. An ER model is a high-level conceptual data model that assists rigorous analysis of the data requirements to build a well-designed database. The ER model depicts real-world entities and the relationships between them [32].

RDF uses *properties* and *property values* to describe resources, which are named by URIs. The three types of objects in RDF are resources, properties, and statements. Each data object in a dataset on the web is considered a resource and is named using a URI. The resources available on the web must be named using HTTPS URIs. The W3C has defined the `rdf:type` property to represent the “object of one or more classes” [13]. Resources are described by their characteristics, called *properties*. The object of RDF that is used to combine a resource with a property of its value is a *statement*. A

statement, in general terms, can be seen as building blocks of a statement in English, which are a subject, a predicate, and an object. An object in a statement can be either a property value, a resource, or a literal. RDF, according to Beckett, Gandon and Schreiber [2, 33], can be represented in three ways: using eXtensible Markup Language (XML) in an RDF/XML document, as triples in subject-predicate-object form or in graphical form.

The University of Manchester has developed Pizza ontology [14] which acts as an inspiration for “pizza” examples in this Chapter. Consider the following statement:

Pizza has CaloricContent with some integer value (2.4)

Figure 2.4 shows the RDF/XML, which consists of various XML tags. An RDF/XML document starts and ends with `<rdf:RDF>` and `</rdf:RDF>` respectively.

```
<rdf:RDF
  "xmlns:rdf=http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  "xmlns:owl="http://www.w3.org/2002/07/owl#"
  "xmlns:pizza="http://www.semanticweb.org/pizzatutorial/ontologies/2020/PizzaTutorial#"
  "xmlns:xsd="http://www.w3.org/2001/XMLSchema#" >
  <owl:Class rdf:about="pizza:Pizza">
    <owl:onProperty rdf:resource="pizza:hasCaloricContent"/>
    <owl:someValuesFrom rdf:resource="xsd:integer"/>
  </owl:Class>
</rdf:RDF>
```

Figure 2.4: Statement 2.4 expressed as RDF/XML. The first few lines define prefixes as a shorthand for the URIs.

Statement 2.5 shows an example of an RDF triple, where the subject (resource) is `pizza:Pizza`; the predicate (property) is `pizza:hasCaloricContent`; and the object (property value) is `xsd:integer`.

`<pizza:Pizza> <pizza:hasCaloricContent> <xsd:integer>` (2.5)

Figure 2.5 shows Statement 2.4 in graphical format. A resource is represented by an oval; a property value is represented by a rectangle; and a property is represented by an arrow, to show that the property (predicate, `pizza:hasCaloricContent`) connects the resource (subject, `pizza:Pizza`) with the value (object, `xsd:integer`).

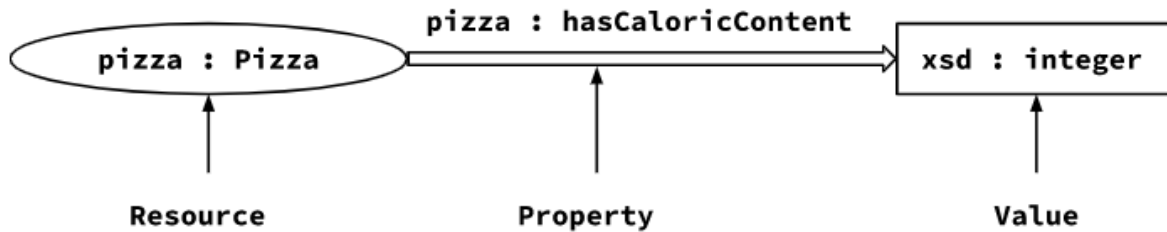


Figure 2.5: Graphical representation of Statement 2.4.

The subject and object of an RDF triple are nodes in a graph and the predicate is an edge that connects the nodes. Considering an entity-relationship data model, the predicate can be treated as a relationship shared by the subject and the object. In addition to the RDF/XML and the line-based N-triples [3] formats, the RDF data model has a variety of serialization formats such as Terse RDF Triple Language (Turtle); Notation3 (N3), which is an extension of N-Triples (line based) and adds a graph label to each line that allows multiple graph encoding [12].

There can be more details added to Statement 2.4 (illustrated in Figures 2.4 and 2.5) and Statement 2.5. That is, more statements can be added in relation to the original statement. Suppose the new statement added is:

`Pizza has PizzaBase` (2.6)

where the property (predicate, `pizza:hasBase`) connects the resource (subject, `pizza:Pizza`) with the value (object, `pizza:PizzaBase`). Figure 2.6 shows a graphical representation of the two example statements combined. To add more complexity to the system, a description of the types of `PizzaBase` can be included. Figure 2.7 shows

a graphical representation of this system.

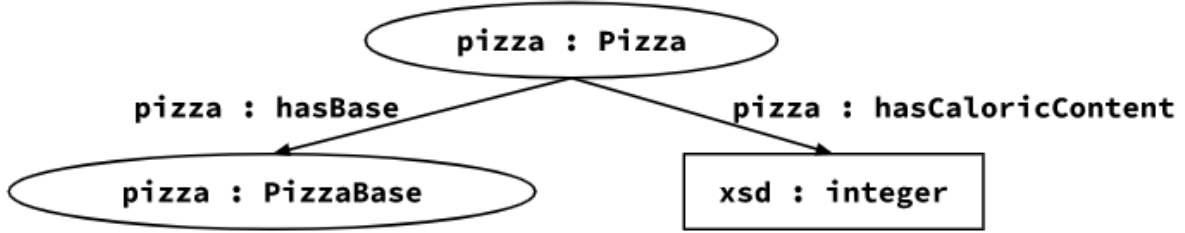


Figure 2.6: Graphical representation of Statements 2.4 and 2.6 combined, following the RDF Primer [34].

PizzaBase has types DeepPanBase and ThinAndCrispyBase (2.7)

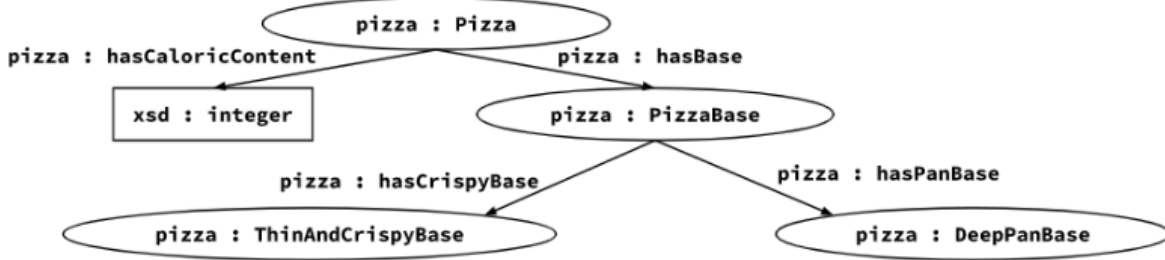


Figure 2.7: Graphical Representation of RDF with types of PizzaBase, after the RDF Primer [34].

`pizza:Pizza` (subject) is connected to `pizza:PizzaBase` (object) node through `pizza:hasBase` (predicate). Further, the `pizza:PizzaBase` node has multiple child nodes, namely, `pizza:ThinAndCrispyBase` and `pizza:DeepPanBase` where the URIs for the predicates are `pizza:hasCrispyBase` and `pizza:hasPanBase` respectively. Instead of using an unnecessary URI for the `pizza:PizzaBase` entity, a blank node can be used to denote the entity, as shown in Figure 2.8. The relationships of the blank node with other nodes give sufficient details without hindering the flow of the data tree.

According to the W3C document *RDF 1.1 Concepts and Abstract Syntax*⁴, the

⁴<https://www.w3.org/TR/rdf11-concepts/>

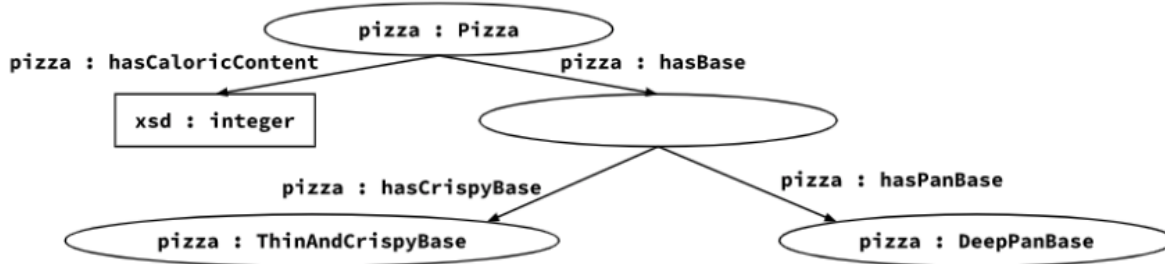


Figure 2.8: RDF Representation in Figure 2.7 without a URI (blank node) for `PizzaBase` Entity, after the RDF Primer [34].

design should “have a simple data model; formal semantics and provable inference; should use an extensible URI-based vocabulary and an XML-based syntax; should support the use of XML schema datatypes; and should allow any one to make statements about any resource”.

Resource Description Framework Schema (RDFS), according to McBride [35], was introduced by the W3C as a technique to define resource types and property names. The data publishers can use RDFS to define various vocabularies in an RDF model. Some of the commonly used RDFS classes and properties⁵ along with their description are listed in Table 2.1 [35].

2.1.3 Open Data

According to the Open Knowledge Foundation [20], “Open Data is data that can be freely used, re-used and redistributed by anyone - subject only, at most, to the requirement to attribute and share-alike”. To make open data available to everyone on the web, the data don’t need to be interlinked. Important features of open data are:

- Availability and access: the data should be available on the web. The users should be able to access the data easily by downloading them over the internet. It should be easy to access and modify the data.

⁵<https://www.w3.org/TR/rdf-schema/>

Class/Property	Description
rdfs:Resource	All things described by RDF are called resources
rdfs:Class	The class of resources that are RDF classes
rdf:Property	The class of RDF properties. It is an instance of rdfs:Class
rdfs:Literal	The class of literal values like integers and strings
rdfs:Datatype	The class of datatypes. All the objects of this class correspond to the RDF model of datatype
rdf:HTML	The class of HTML literal values and subclass of rdfs:Literal
rdfs:range	An instance of rdf:Property that is used to state that the values of a property are instances of one or more classes
rdfs:domain	An instance of rdf:Property that is used to state that any resource that has a given property is an instance of one or more classes
rdf:type	An instance of rdf:Property that is used to state that a resource is an instance of a class
rdfs:subClassOf	An instance of rdf:Property that is used to state that all the instances of one class are instances of another
rdfs:label	an instance of rdf:Property that may be used to provide a human-readable version of a resource's name
rdfs:subPropertyOf	An instance of rdf:Property that is used to state that all resources related by one property are also related by another

Table 2.1: Commonly used Classes and Properties of RDFS

- Re-use and redistribution: the data on the web must have permissions to re-use and redistribute in combination to other datasets.
- Universal participation: the data should be available to use, re-use, and redistribute for all the users without any discrimination to an individual or a group.

Linked data is concerned with publishing structured data, along with the semantics of the relationships amongst that data, onto the web. The W3C standards and technologies handle the distribution and engagement of this data across the web with the potential to present it as one huge database [5].

2.1.4 Wikidata

A variety of projects and websites, that use linked open data, have been developed. Wikidata is one of the major projects of linked open data that present structured data on the semantic web.

Krotzsch and Vrandečić [41] describe the introduction of Wikidata to the world in 2012. According to Erxleben et al. [17], Wikidata is “the community-created knowledge base of Wikipedia and the central data management platform for Wikipedia”. It was established for the collection and integration of Wikipedia data into the semantic web. Wikidata is a free and open-source tool that handles a large amount of structured data and maintains pages that store the data. Each entity on Wikidata represents the subject of a triple with a dedicated page. The properties of Wikidata are similar to that of RDF where the individuals and classes are indicated using items. Data on a page of the Wikidata is available for the users to access as well as edit.

For instance, the University of Regina (Q3104287) has an item page⁶. Figure 2.9 illustrates a portion of the contents of this page.

Wikidata uses a method of automatically assigning unique identifiers to the entities, which are used as the labels of the item pages instead of the actual names of the entities. These identifiers are a string starting with a capital Q followed by a sequence of digits. The identifiers are assigned irrespective of the language as the Wikidata website supports multiple languages.

An item page on Wikidata consists of two main sections. The first section, entitled Statements, contains a list properties and values. The properties, along with their property IDs, used for the University of Regina are listed in Table 2.2. The other section is Identifiers, which consists of various IDs associated with the subject. The identifiers, along with their property IDs, used for the University of Regina are listed in

⁶<https://www.wikidata.org/wiki/Q3104287>

university in Saskatchewan, Canada

Regina College | Regina Campus of the University of Saskatchewan

[In more languages](#)
[Configure](#)

Language	Label	Description	Also known as
English	University of Regina	university in Saskatchewan, Canada	Regina College Regina Campus of the Universit...
Canadian English	No label defined	No description defined	
French	université de Regina	No description defined	Université de Régina Universite de Regina Université de regina
Italian	No label defined	No description defined	

All entered languages

Instance of	Value	Actions
university	- 0 references	edit + add reference
	open-access publisher	edit - 1 reference stated in Directory of Open Access Journals + add reference + add value

image

edit



University of Regina Library viewed from the Oval.jpg
3,222 × 2,250; 662 KB

- 0 references

+ add reference

+ add value

Inception **1911** [edit](#)

– 1 reference

imported from Wikimedia project [French Wikipedia](#)

Figure 2.9: A portion of the item page for University of Regina on Wikidata.

Table 2.3 where as the RDF/XML description of the University of Regina on Wikidata is as shown in Figure 2.10.

Statement Property	Property ID
instance of	P31
image	P18
inception	P571
country	P17
located in the administrative territorial entity	P131
coordinate location	P625
subsidiary	P355
official website	P856
commons category	P373
topic's main category	P910
category for employees of the organization	P4195
category for alumni of educational institution	P3876
API endpoint	P6269
social media followers	P8687

Table 2.2: Statement Properties and Property IDs for `University of Regina` on Wikidata.

2.2 Ontologies

According to Gruber [24], ontology is an `explicit specification of a conceptualization`, where conceptualization is “an abstract, simplified view of the world (includes objects, concepts and other entities that are assumed to exist in some area of interest and the relationships that hold among them) that we wish to represent for some purpose”. In web semantics, an ontology is used to define the semantics of the resources briefly and methodically. On the semantic web, the ontologies have been used to define resources’ conceptual and physical qualities in the form of metadata schema for a predetermined group of users.

Identifier Property	Property ID
VIAF ID	P214
ISNI	P213
GND ID	P227
Library of Congress authority ID	P244
WorldCat Identities	P7859
ARWU university ID	P5242
Crossref funder ID	P3153
Facebook ID	P2013
Freebase ID	P646
Google Maps Customer ID	P3749
GRID ID	P2427
HAL structure ID	P6773
LittleSis organization ID	P3393
Microsoft Academic ID	P6366
Ringgold ID	P3500
ROR ID	P6782
Times Higher Education World University ID	P5586
Twitter username	P2002
U-Multirank university ID	P5600

Table 2.3: Identifier Properties and Property IDs for **University of Regina** on Wikidata.

2.2.1 Web Ontology Language

The W3C developed a standard ontology language called OWL⁷ to define and describe concepts for ontology using different facilities and operations like intersection, union and negation. It is based on a logical model on which a reasoner can be used to check the mutual consistency among the statements and definitions in the ontology. The reasoner is also responsible for the identification of a suitable concept for particular definitions [28].

⁷<https://www.w3.org/TR/owl-overview/>

```

<?xml version="1.0" encoding="utf-8" ?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema#"
  xmlns:ontolex="http://www.w3.org/ns/lemon/ontolex#"
  xmlns:wdt="http://www.wikidata.org/prop/direct/"
  xmlns:wdrn="http://www.wikidata.org/prop/direct-normalized/">
  <rdf:Description rdf:about="https://www.wikidata.org/wiki/Special:EntityData/Q3104287">
    <rdf:type rdf:resource="http://schema.org/Dataset"/>
    <schema:about rdf:resource="https://www.wikidata.org/entity/Q3104287"/>
    <cc:license rdf:resource="http://creativecommons.org/publicdomain/zero/1.0/" />
    <rdf:Description rdf:about="https://en.wikipedia.org/wiki/University_of_Regina">
      <rdf:type rdf:resource="http://schema.org/Article"/>
      <schema:about rdf:resource="https://www.wikidata.org/entity/Q3104287"/>
      <schema:inLanguage>en</schema:inLanguage>
      <schema:isPartOf rdf:resource="https://en.wikipedia.org/" />
      <schema:name xml:lang="en">Univeristy of Regina</schema:name>
      <schema:description xml:lang="en">university in Saskatchewan, Canada</schema:description>
      <wdt:P18 rdf:resource="http://commons.wikimedia.org/wiki/Special:
        FilePath/University%20of%20Regina%20Library%20viewed%20from%20the%20val.jpg"/>
    </rdf:Description>
  </rdf:Description>
</rdf:RDF>

```

Figure 2.10: Wikidata Entry (RDF/XML) for University of Regina.

2.2.2 Components of OWL Ontologies

An OWL ontology consists of a set of components that define and describe the concepts collectively. The basic components of any OWL ontology are briefly explained in this Section.

2.2.2.1 Individuals

According to Horridge et al. [28], “Individuals represent objects in the domain in which we are interested”. Individuals are the objects of classes. Figure 2.11 shows the representation of different individuals in various domains. The triangles are used here to denote the individuals in free space.

2.2.2.2 Properties

Binary relationships, links shared by two individuals (things), are called properties. For instance, the individuals `Customer1` and `AmericanaPizza1` from Figure 2.11 are linked to each other by the property `purchasedPizza`. Each property can have its

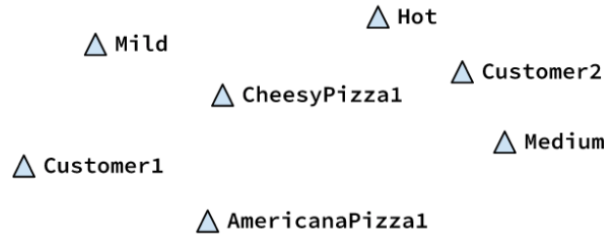


Figure 2.11: Representation of individuals in OWL.

inverse. For instance, `purchasedPizza` can have an inverse property `purchasedBy`. The properties can be symmetric or transitive. It is possible to have functional properties where the properties can be restricted to having only one value.

Figure 2.12 shows representation of properties between the individuals `Customer1` and `Medium`, and `Customer1` and `AmericanaPizza1`. The relation between `Customer1` and `Medium` is `hasSpicinessPreference`. The arrow between two individuals in the figure denotes the property (relationship) between them.

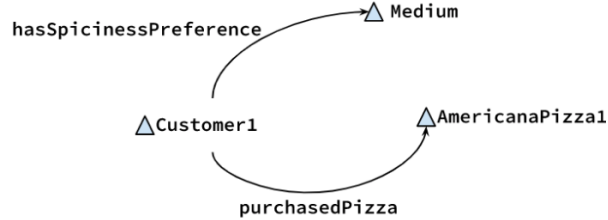


Figure 2.12: Representation of properties and individuals in OWL.

2.2.2.3 Classes

In OWL, the sets consisting of individuals are called classes. Mathematical (formal) descriptions are used to describe the OWL classes. These descriptions include accurate statements on the requirements for the class membership. For instance, the class `Customer` consists of all the individuals that are customers in the Domain of Discourse which can contain one or more classes. Inheritance in OWL classes (Class Hierarchy) may exist with the classes having subclasses and/or super-classes which

is technically termed as taxonomy. “Subclasses specialize (“are subsumed by”) their super-classes” [28]. For instance, consider the classes **Person** and **Customer** from the pizza ontology, where **Customer** is a subclass of **Person**. This implies the following statements:

- All customers are persons
- All members of class **Customer** are members of class **Person**
- Being a **Customer** implies that individual is a **Person**
- **Customer** is subsumed by **Person**

Figure 2.13 shows the representation of the classes for the individuals and properties that are shown in Figure 2.12.

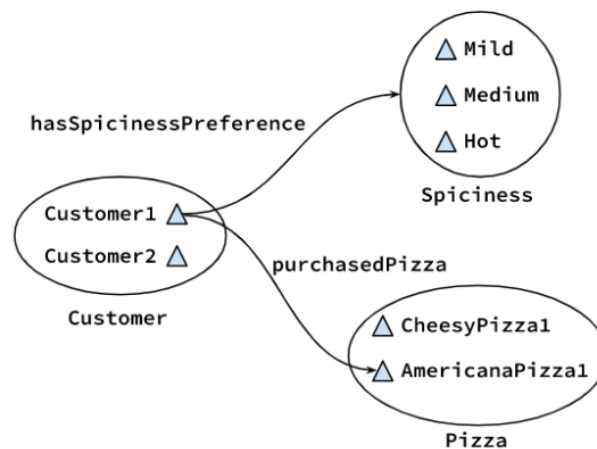


Figure 2.13: Representation of classes, individuals, and properties in OWL.

2.2.3 Ontology Editors

Ontology editors are used by data publishers to create new ontologies when existing vocabularies are not suitable for their purpose. Of the ontology editors listed on

the W3C wiki page⁸ that appear to be active with a current cross-platform application available (Protégé, Neologism, TopBraid Composer, Anzo, Semantic Turkey, and VocBench), Protégé [23] is the most popular amongst researchers. Protégé also has a well-established tutorial [14] explaining its use.

2.3 Linked Data Life Cycle

The Linked Data Life Cycle described by Ngomo et al. [36] includes multiple stages, which are listed in Table 2.4. It is not required to go through all the stages sequentially to publish LOD on the web. The first stage is extraction of non-RDF data and conversion of that data into RDF. Then, any or all of the other stages listed in Table 2.4 can be performed.

Stage	Description
Extraction	Extraction and conversion of unstructured and semi-structured data into RDF
Storage & Querying	Storing the RDF data in a format that supports data querying efficiently
Authoring	Allowing the users to create new data or to access and modify the existing data
Linking	Creating links among the data belonging to various domains and users based on their entities
Enrichment	Data enrichment for efficient query support using various top-tier structures
Quality Analysis	Managing data using different strategies to ensure good quality of available data on the web
Browsing & Exploration	Making the structured data available on the web for the users to browse and explore effectively

Table 2.4: Stages in the Linked Data Life Cycle described by Ngomo et al. [36]

⁸https://www.w3.org/wiki/Ontology_editors

2.4 Mapping of Data from Spreadsheets to RDF

The preferred format of the data published on the semantic web is an RDF format. Many scientists store and publish data using CSV, rather than RDF, format because conversion of data to RDF is comparatively difficult. A number of open source tools [16], listed on the W3C wiki⁹, can be used to do the conversion. Two worth highlighting are OpenRefine and RML Mapper.

OpenRefine¹⁰, is a Java-based set of tools that work with messy data using Google Refine Expression Language (GREL) It allows users to extract, transform, export, and convert CSV, Excel, spreadsheets and other tabular data into RDF where a GUI (Graphical User Interface) is used to define schema mapping. The RDF mapping is accomplished with an extension.

“An RDF Mapping Language (RML) mapping defines a mapping from any logical source to RDF. It is a structure that consists of one or more triples maps.”¹¹ The RML rules can be written and tested in YARRRML (a more accessible way to use RML) using tools like the Matey Web UI¹².

Once the RML rules have been defined, the next step is generate RDF from the RML mappings.

RMLMapper¹³ is a reference implementation written in Java. It supports custom functions and operations with small files as it loads all data in memory.

⁹<https://www.w3.org/wiki/ConverterToRdf>

¹⁰<https://openrefine.org>

¹¹<https://rml.io/specs/rml/#example-XML>

¹²<https://rml.io/yarrml/matey/>

¹³<https://github.com/RMLio/rmlmapper-java>

2.5 Summary

The building blocks discussed here are used to define a generic process for publishing LOD from spreadsheet data (Chapter 3), which is then applied to a real example with Hepting's student feedback data (Chapter 4).

Chapter 3

Methodology

This Chapter focuses on the methodology that is required to create and publish LOD. Chapter 4 presents a proof of concept for this methodology with real data.

3.1 Describe Semantics

It is feasible to use existing linked open vocabularies to develop new ontologies [31]. “Linked Open Vocabularies” available at <https://lov.linkeddata.es/dataset/lov> serves as the main resource to linked open vocabularies. There exist some vocabularies for “Survey” but none of them describes “Likert” items.

Data publishers will have to create a vocabulary in case they do not find the required vocabulary from those already existing.

3.2 Describing New OWL Vocabularies

There are editors available that help the publishers to create their own linked open vocabularies. The editors can be compared and chosen based on some basic features like *General Description* (developers’ details, and accessibility on the internet), *Software Architecture and Tools Evaluation* (semantic web architecture, support for plug-

ins, backup management, and ontology storage), *Interoperability* (interoperability with other ontology tools, support and translation for other languages) *Knowledge Representation* (Knowledge Representation paradigm of knowledge model, axiom language, and methodological support), *Inference Services* (built-in inference engine, other attached inference engines, and constraint/consistency checking), and *Usability* (graphical taxonomy, graphical views, zooms, collaborative working, and ontology libraries).

After studying the details and comparison among some popular editors by Clunis and Sashanie [11], Protégé has been used for the development of the instrument to support this research and thesis work as it is the most popular amongst researchers.

3.2.1 Protégé

Horridge et al. [23] have developed Protégé¹, “a free, open-source ontology editor and framework for building intelligent systems”.

3.2.1.1 Usability of Protégé

The International Organization for Standardization (ISO) standard 9241² defines usability as “the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use”. In usability engineering, Jacob Neilson [37] suggested five qualities of a usable product: **learnability**, **efficiency**, **memorability**, **errors** (low rate, easy to recover), and **satisfaction**. These were the original five dimensions of usability which were then re-termed as all five words starting with “E” by Quesenbery [39], and they became popular as the five Es of usability. The new terms are *Effective*, *Efficient*, *Engaging*, *Error-tolerant*, and *Easy-to-learn*.

¹<https://protege.stanford.edu>

²<https://www.iso.org/standard/63500.html>

The five dimensions of usability can be considered while designing as well as for evaluation. While designing, the engineers consider the five dimensions as goals to be achieved whereas while testing and evaluation, they are considered as the requirements necessary for the designed model. The five dimensions of usability were used to assess the usability of Protégé and the report of the evaluation is as follows:

- Effective: The task is completed accurately

Protégé is an effective design as it allows the users to build ontologies and use other tools precisely. The users can complete the desired tasks accurately using the set of facilities available.

- Efficient: The task is completed quickly

It is easy and straightforward to describe the vocabularies and build ontologies. All the desired tasks in Protégé can be completed quickly (without wasting much time) as it is easy to look for the facilities using the menus and tools available. Hence, Protégé is an efficient ontology builder.

- Engaging: The look and feel of the design is interactive and engaging

The tools and menus in Protégé are interactive which leads to a better experience for the user. Also, the layout and the components are attractive and decent for the user to use this ontology builder whenever necessary. Overall, Protégé gives an engaging and satisfactory user experience.

- Error-tolerant: Error prevention and data recovery in case any error is committed

Protégé provides options for modifications, additions, and deletions of classes, objects, and properties, which allow the users to correct the errors that were committed. Also, the deleted classes can be recovered using **Undo** from the edit menu resulting in proving Protégé as an error-tolerant design.

- Easy-to-learn: The interface should be easy to learn

Protégé is easy to learn as it has a familiar look and commonly used names for the tools and menus making it easy for the users to adapt and know them nicely. Also, the help documentation along with a list of some frequently asked questions is available, which helps the users in the correct direction whenever they are lost.

3.2.2 Procedure to Describe OWL Vocabularies Using Protégé

This Section focuses on a step-wise methodology that has been followed to create new OWL vocabularies using Protégé, which is available for use as a web application or for download in the case of the desktop application.

To commence with the process of building an ontology, a new project needs to be created with a unique Internationalized Resource Identifier (IRI)³ and the ontology version.

3.2.2.1 Classes and Class Hierarchy

The next step is to create classes and build the hierarchy for the class nodes. Each entity in Protégé is directly or indirectly a child node of the `Thing` class. Also, each object has a name-value pair, which generally are the nodes of the `Domain_Entity` class, an inherited class of `Thing`. Initially, four classes, `Thing`, `Independent_Entity`, `Domain_Entity`, and `Value` are created using class hierarchy, as shown in Figure 3.1.

To create a class hierarchy for the classes that represent entities in an ontology, there are three steps to be followed:

1. Select `owl:Thing` in the Class Hierarchy View of the Classes tab from the Windows menu followed by selecting `Create class hierarchy...` from the Tools

³<https://www.w3.org/2011/rdf-wg/wiki/IRIs/RDFConceptsProposal>

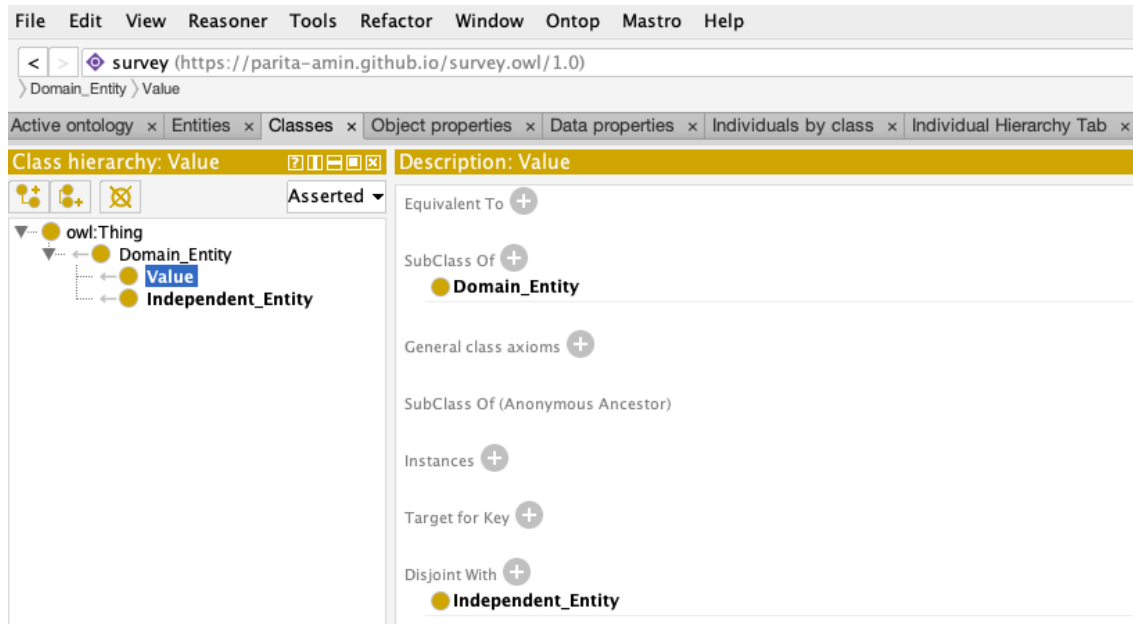
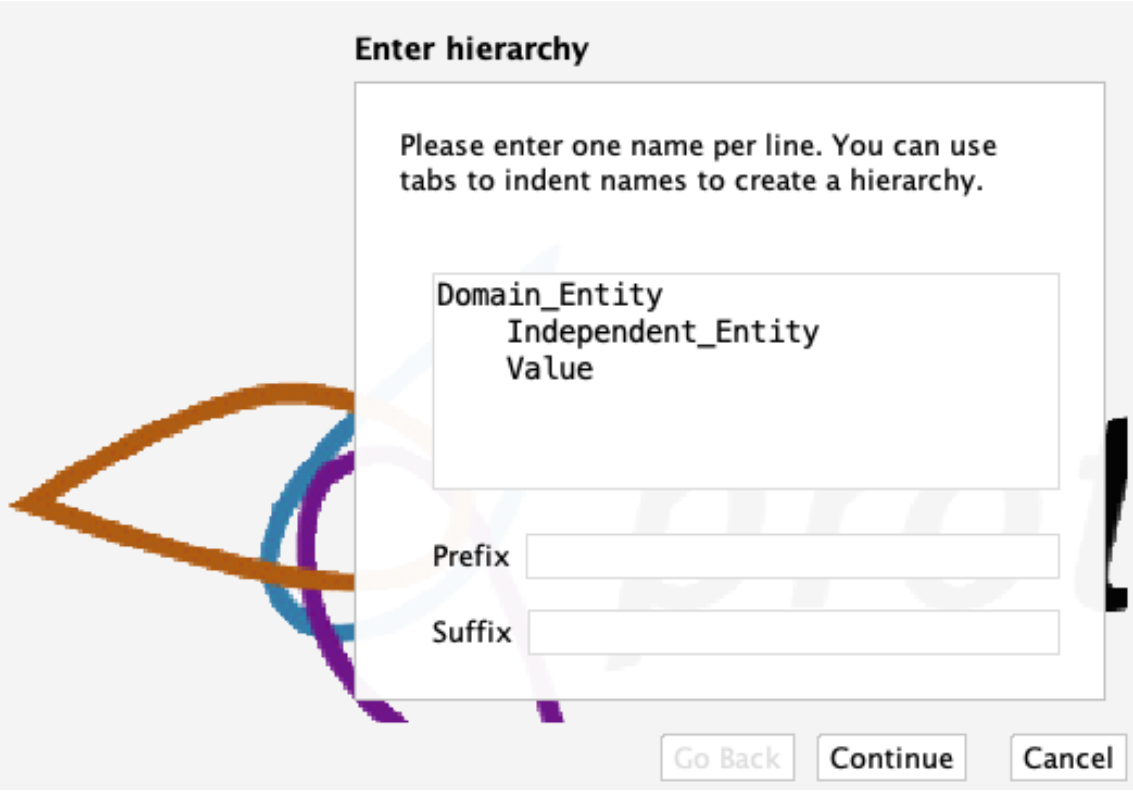


Figure 3.1: Class Hierarchy of Thing, Domain_Entity, Independent_Entity, and Value

menu, to open a **Enter hierarchy** pop-up window which allows the users to add classes to the ontology.

2. Enter the class names in the pop-up window starting from the left. For the child nodes, leave some spaces before writing the class name. Each horizontal line consists of one class name, as shown in Figure 3.2.
3. On clicking **Continue** on the pop-up window in Figure 3.2, it opens up a new pop-up window. Here, it is important to state whether one wants the sibling classes to be disjoint or not. The checkbox needs to be checked accordingly. Usually, it is recommended to create disjoint sibling classes which can be changed later as necessary.

Once the classes are added, the next step is to describe the classes. Descriptions of classes like **Equivalent to**, **SubClass Of**, **Instances**, **Disjoint With** and so on can be added using the **Description** view of the **Class** tab (shown in Figure 3.1). Each class



The image shows a 'Pop-up Window' titled 'Enter hierarchy'. The window has a light gray border and a white background. At the top, the title 'Enter hierarchy' is displayed in bold black text. Below the title, a message reads: 'Please enter one name per line. You can use tabs to indent names to create a hierarchy.' This message is followed by a text area containing three lines of text: 'Domain_Entity', 'Independent_Entity', and 'Value'. The text 'Independent_Entity' is indented under 'Domain_Entity', and 'Value' is indented under 'Independent_Entity'. Below the text area, there are two input fields: 'Prefix' and 'Suffix', each with a corresponding text box. At the bottom of the window, there are three buttons: 'Go Back', 'Continue', and 'Cancel'. The window is overlaid on a background that features a large, stylized, colorful graphic on the left side, consisting of several overlapping curved lines in orange, blue, and purple.

Enter hierarchy

Please enter one name per line. You can use tabs to indent names to create a hierarchy.

Domain_Entity
Independent_Entity
Value

Prefix

Suffix

Figure 3.2: Enter hierarchy Pop-up Window

can be described individually by selecting a particular class and adding details to the description view.

3.2.2.2 Object Properties and Object Property Hierarchy

The next step in building the ontology, after describing the classes, is describing the `Object Properties`. Object properties are used to link the entities (classes in Protégé). The `owl:topObjectProperty` is the root node of the object properties' hierarchy. The hierarchy for object properties can be built by selecting a property in the `Object Property Hierarchy` view of the `Object Property Views` tab and then going to the tools menu and selecting `Create object property hierarchy....` The remaining steps are similar to the ones that are followed for building class hierarchy (Section 3.2.2.1). Also, the description view of object properties hierarchy allows the user to describe the object properties. The details that can be added or modified using the description view are `Equivalent To`, `SubProperty Of`, `Inverse Of`, `Domains (intersection)`, `Ranges (intersection)`, `Disjoint With`, and `SuperProperty Of (Chain)`.

The names of the object properties usually start with an `is` or a `has`. For instance, consider two classes from the pizza ontology: `Pizza` (class 1) and `PizzaBase` (class 2). To describe the link between class 1 (`Pizza`) and class 2 (`PizzaBase`) such that class 1 has value class 2 and class 2 is value of class 1, the properties `hasBase` and `isBaseOf` would be used respectively, as shown in Figure 3.3.

Each object property might have a corresponding inverse property. For the above instance, it can be said that:

`Pizza hasBase PizzaBase is equivalent to PizzaBase isBaseOf Pizza`

Here, `hasBase` is inverse of `isBaseOf` and `isBaseOf` is inverse of `hasBase`. Apart from inverse, domain and range⁴ are two other important parts of description of object

⁴<http://protegeproject.github.io/protege/views/object-property-description/>

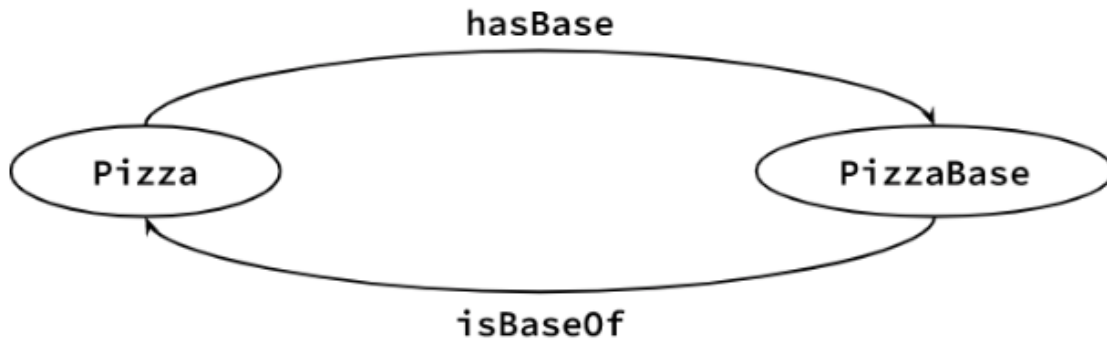


Figure 3.3: Object Properties for Classes **Pizza** and **PizzaBase**

properties. According to Horridge et al. [28], in OWL, “domain and range are not constraints to be checked. They are axioms which are used by the reasoner to make inferences”. As shown in Figure 3.4, **Pizza** acts as the domain and **PizzaBase** acts as the range for **hasBase** whereas for **isBaseOf**, **PizzaBase** is the domain and **Pizza** is the range.

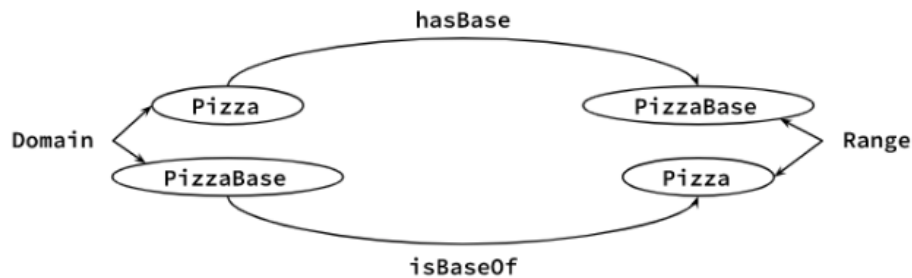


Figure 3.4: Domain and Range Classes for **hasBase** and **isBaseOf**

3.2.2.3 Data Properties and Data Property Hierarchy

Furthermore, the **Data Properties** can also be described for ontology. “An ontology data property provides a relation to attach an entity instance to some literal datatype value (an RDF number, string or data for example) that is a measure or estimate of what that data property is about.”⁵

⁵<https://ddooley.github.io/docs/data-properties/>

In Protégé, the `owl:topDataProperty` is the root node of the data properties' hierarchy. The data property hierarchy can be built by selecting a property in the Data Property Hierarchy view of the Data Property Views tab and then going to the tools menu and selecting **Create data property hierarchy....** The remaining steps are similar to the ones that are followed for building class and object property hierarchy (Section 3.2.2.1). Data properties can be described using the description view of data properties of hierarchy which included details like **Equivalent To**, **SubProperty Of**, **Domains (intersection)**, **Ranges**, and **Disjoint With**.

3.2.2.4 OWL Ontology Inconsistencies and Reasoner

After describing the classes, object properties, data properties, and individuals, the next step is to determine and fix the errors and inconsistencies in the ontology using a Reasoner. The current version of the Protégé IDE (Integrated Development Environment) offers support for some built-in semantic reasoners which are listed below:

- ELK 0.4.3⁶ : A Java-based fast reasoner for lightweight OWL2 EL which is available under Apache License 2.0
- FaCT++ 1.6.5⁷ : A Tableaux-based reasoner for expressive OWL Description Logics (DL) which is implemented using C++
- HermiT 1.4.3.456⁸ : A Java-based reasoner, written using OWL, that can be used to determine consistency of the ontology as well as for the identification of relationships among the classes of the ontology
- Mastro DL-Lite⁹ : An OBDA (Ontology-Based Data Access) management system

⁶<https://www.w3.org/2001/sw/wiki/ELK>

⁷<https://www.w3.org/2001/sw/wiki/Fact>

⁸<https://www.w3.org/2001/sw/wiki/Hermit>

⁹https://protegewiki.stanford.edu/wiki/Mastro_DL-Lite_Reasoner

in which the DL-Lite family of languages for lightweight DL are used for ontology specifications

- Ontop 4.1.0¹⁰ : A reasoner which uses SPARQL for querying datasets as Virtual RDF Graphs
- Pellet and Pellet (incremental)¹¹ : A Java-based OWL2 reasoner for the optimization of nominals, conjunctive query answering, and incremental reasoning

3.2.2.5 The FAIR Principles

After developing the linked data vocabulary, the next step is to use this vocabulary, either to develop another ontology or to map data but before that, it is important to check if the developed vocabulary meets the standards. The FAIR principles [42] for linked open data were published in 2016. According to the GoFAIR [22], “the authors intended to provide guidelines to improve the Findability, Accessibility, Interoperability, and Reuse of digital assets”. The FAIR principles as listed by GoFAIR [22] are:

- Findable: The metadata and data should be easy to find for both humans and computers in order to be reused.
- Accessible: After finding the data, the data needs to be accessible, possibly including authentication and authorization.
- Interoperable: Users should be able to integrate the data with other data, and interoperate with applications or workflows for analysis, storage, and processing.
- Reusable: To optimize the reuse of the data, metadata and data should be well-described so that they can be replicated and/or combined in different settings.

¹⁰<https://www.w3.org/2001/sw/wiki/OnTop>

¹¹<https://www.w3.org/2001/sw/wiki/Pellet>

Chapter 4 discusses the results of using FAIR principles to evaluate the **Survey** vocabulary.

3.2.3 Data Validation

Hepting collects feedback responses and manually stores them in spreadsheets. Much of this data is collected on paper, with a handful of spreadsheets created directly from within the University of Regina’s learning management system. It is important to validate the stored data so that the data only consists of acceptable values. According to Ekaputra and Xiashuo [15], “One of the challenges of adopting Semantic Web Technologies (SWT) for enterprise data integration is to provide the means to define and validate structural constraints over RDF graphs.” To address this challenge, they developed SHACL4P, a Protégé plugin. SHACL4P is used for defining and validating SHApes Constraint Language (SHACL), which is a W3C standard for the validation of contents of an RDF-style graph database¹². SHACL uses **shapes** to describe constraints.

3.3 Mapping Tabular Data to RDF

After describing the best possible version of the ontology, the next step is to convert the data stored in spreadsheets into linked data using either OpenRefine or RMLMapper.

¹²<https://www.w3.org/TR/shacl/>

3.3.1 Mapping Using OpenRefine

OpenRefine is a free, open-source powerful tool for working with messy data. OpenRefine can be downloaded and installed from the OpenRefine website¹³; run as a web server; and accessed using local web server in the browser. The first step is to load OpenRefine and access it in the browser.

OpenRefine offers options to create a new project as well as to open an existing project. The users can open an existing project using **Open Project** and continue working on the data. Creating a new project includes following steps:

1. Click on **Create Project**, select the type on sources of dataset and then click on **Choose Files** button to select data sources. OpenRefine supports imports from TSV, CSV, JSON, XML, RDF/XML, Excel, and Google Data documents whereas the other file types can be supported using OpenRefine extensions.
2. The next step is to decide the orientation of the data to be displayed using various options as shown in Figure 3.5. The **Update Preview** button allows viewing the results of these modifications.

The screenshot shows the 'Parse data as' dialog in OpenRefine. On the left, a sidebar lists file types: 'CSV / TSV / separator-based files' (selected), 'Line-based text files', 'Fixed-width field text files', 'PC-Axis text files', 'JSON files', 'MARC files', 'JSON-LD files', 'RDF/N3 files', 'RDF/N-Triples files', and 'RDF/Turtle files'. The main area has 'Character encoding' set to 'UTF-8'. Under 'Columns are separated by', 'commas (CSV)' is selected. There are checkboxes for 'Ignore first', 'Parse next' (checked), 'Discard initial', 'Load at most', and 'Use character' (checked). To the right of these are input fields for 'line(s) at beginning of file' (0), 'line(s) as column headers' (1), 'row(s) of data' (0), and 'row(s) of data' (0). Below these are checkboxes for 'Attempt to parse cell text into numbers' and 'Store blank rows' (checked). At the bottom right are checkboxes for 'Store blank cells as nulls', 'Store file source', and 'Store archive file'. An 'Update Preview' button is in the top right corner.

Figure 3.5: Options to Set Orientation of Data in New OpenRefine Project

3. The final step is to mention the name of the project and click on the **Create Project »** button as shown in Figure 3.6.

OpenRefine provides support for following tasks:

¹³<https://openrefine.org/>

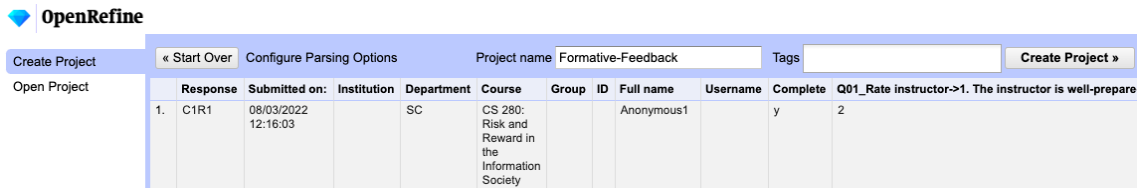


Figure 3.6: Naming the New OpenRefine Project

1. Exploring Data:

Large data sets can be explored with ease using OpenRefine. The **Facet** option allows users to explore values of particular type by offering options like **Text facet**, **Numeric facet**, **Timeline facet**, **Scatterplot facet**, and other custom facet. Figure 3.7 shows the facet options offered by OpenRefine for a field.

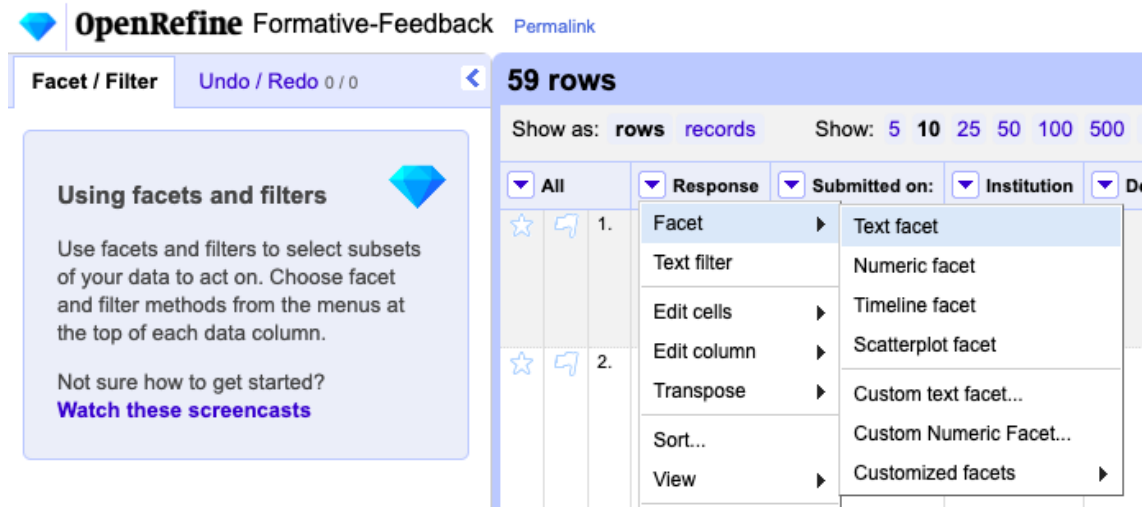


Figure 3.7: Exploring Data in OpenRefine - 1 of 3

The left side of the screen in Figure 3.8 shows the result of using **Text facet** for **Response** column. It also offers an option to **edit** and **include** the results of the facet. Figure 3.9 shows the result of using **include** for the record corresponding to the response C1R1.

2. Cleaning and Transforming Data:

OpenRefine Formative-Feedback [Permalink](#) Open... Export Help

Facet / Filter Undo / Redo 0 / 0 <

Refresh Reset All Remove All

Response change

59 choices Sort by: name count Cluster

C1R1 1
C1R10 1
C1R11 1
C1R12 1
C1R13 1
C1R14 1
C1R15 1
C1R16 1
C1R17 1
C1R18 1
C1R19 1
C1R2 1
C1R20 1
C1R21 1

59 rows Extensions: RDF Wikidata

Show as: rows records Show: 5 10 25 50 100 500 1000 rows
« first < previous 1 of 6 pages next > last »

			Response	Submitted on:	Institution	Department	
☆	1.		C1R1	08/03/2022 12:16:03		SC	CS Risk and Reward in the Information Society
☆	2.		C1R2	03/03/2022 14:42:35		SC	CS Risk and Reward in the Information Society
☆	3.		C1R3	07/03/2022 10:15:14		SC	CS Risk and Reward in the Information Society

Figure 3.8: Exploring Data in OpenRefine - 2 of 3

OpenRefine Formative-Feedback [Permalink](#) Open... Export Help

Facet / Filter Undo / Redo 0 / 0 <

Refresh Reset All Remove All

Response change invert reset

59 choices Sort by: name count Cluster exclude

C1R1 1
C1R10 1
C1R11 1
C1R12 1

1 matching rows (59 total) Extensions: RDF Wikidata

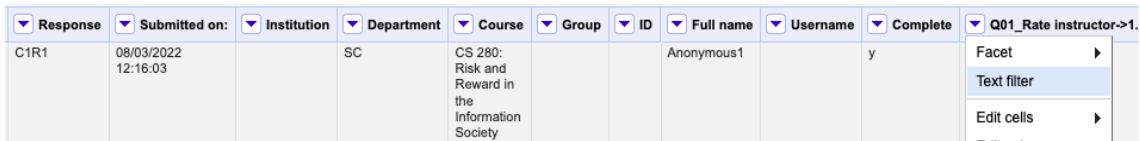
Show as: rows records Show: 5 10 25 50 100 500 1000 rows
« first < previous 1 of 1 page next > last »

			Response	Submitted on:	Institution	Department	Course
☆	1.		C1R1	08/03/2022 12:16:03		SC	CS 280: Risk and Reward in the Information Society

Figure 3.9: Exploring Data in OpenRefine - 3 of 3

- Cleaning Data:

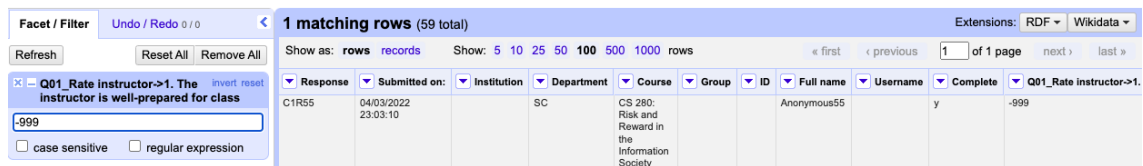
While dealing with messy data, OpenRefine can be used to clean the data using the **Text filter** option. Figure 3.10 shows the **Text filter** option offered by OpenRefine for a field.



Response	Submitted on	Institution	Department	Course	Group	ID	Full name	Username	Complete	Q01_Rate instructor->1
C1R1	08/03/2022 12:16:03		SC	CS 280: Risk and Reward in the Information Society			Anonymous1		y	Facet Text filter Edit cells

Figure 3.10: Cleaning Data in OpenRefine - 1 of 2

Figure 3.11 shows the result of applying the **Text filter** option for the **Q01_Rate instructor->1** field and using **-999** as argument.



Response	Submitted on	Institution	Department	Course	Group	ID	Full name	Username	Complete	Q01_Rate instructor->1
C1R55	04/03/2022 23:03:10		SC	CS 280: Risk and Reward in the Information Society			Anonymous55		y	-999

Figure 3.11: Cleaning Data in OpenRefine - 2 of 2

- Transforming Data:

OpenRefine allows transforming data from one format to another using **Edit cells** option under which **Transform...** and **Common transform** offers different options for transforming data.

3. Reconciling and Matching Data:

OpenRefine can be used to link and extend dataset with various web services. It also supports services (using extensions and plugins) that can be used to upload the cleaned data to central databases like Wikidata. The data can be converted to linked data (as in this research) which can be then uploaded to central databases.

3.4 Announcing Linked Open Data

After the data is converted into linked data, it can be made available on the web in two steps, licensing the data and publishing on the web, which are discussed in this section.

3.4.1 Licensing Linked Data

W3C [29] advises that it is critical to identify the owner of any data published on the web and to explicitly link the licence to the data itself. It is simpler to attach a licence that goes with the data itself when publishing linked open data. When data has a clear, agreeable licence attached to it, people are more likely to reuse it.

The Creative Commons¹⁴ website is a useful resource for open data providers. The legal and technical framework for publishing digital content is developed, supported, and maintained by Creative Commons. There are different licenses available like **Attribution-ShareAlike 4.0 International**¹⁵

3.4.2 Publishing Linked Open Data

According to Petrou et al. [38], the linked data principles state that dereferencable URIs are the preferred method of accessing linked data and that the data should be openly available. When accessed, these URIs offer accurate descriptions of the concepts they stand for in a number of formats. RDF dumps and SPARQL endpoints are hence two ways that users can access the data. Dereferencing of the data is accomplished using the specialised RDF store, and all the data are accessible via a SPARQL endpoint service or through the faceted browsing feature, which allows users to search resources and move between them.

¹⁴<https://creativecommons.org/>

¹⁵<https://creativecommons.org/licenses/by-sa/4.0/>

Publishing datasets to international or national open data portals (e.g. Licence-Canada¹⁶) might further *announce* them to the public and make them discoverable.

¹⁶<https://open.canada.ca/en/open-government-licence-canada>

Chapter 4

Publishing LOD from a Spreadsheet

This Chapter provides a practical application of the methodology presented in Chapter 3 to describe and publish linked open data for a feedback survey. The organization of this Chapter mirrors the preceding one, as the methodology is applied to publish LOD from Hepting’s feedback survey.

4.1 Describing Semantics for Survey Ontology

As described in Chapter 3, to create a new ontology, an existing vocabulary can be reused or a new vocabulary can be described. According to the **Linked Open Vocabularies** website¹, there are 6 survey-related vocabularies described, as shown in Figure 4.1.

Some terms from the **disco-DDI-RDF Discovery Vocabulary** [25] can be reused to describe Dr. Hepting’s student feedback form. For now, there is no existing vocabulary to describe Likert items, so the vocabulary for **Survey** ontology has been described without using any existing vocabularies. The following Section focuses on describing new vocabulary for **Survey** ontology.

¹<https://lov.linkeddata.es/dataset/lov/vocabs?q=survey>

The screenshot shows the Linked Open Vocabularies Website interface. At the top, there are navigation links: VOCABS, TERMS, AGENTS, and SPARQL/DUMP. Below this is a search bar with the word 'survey' entered. The search results are displayed in a list format, showing 6 results. The results are as follows:

Results	Vocabulary Name	URL	Description
6 results	osgeom - Ordnance Survey Geometry Ontology	http://data.ordnancesurvey.co.uk/ontology/geometry/	A provisional small geometry vocabulary used by Ordnance Survey vocabularies and datasets. Likely to change when some standard way of representing geometries in RDF is agreed. @en
	sur - The Survey Ontology	https://www.w3id.org/survey-ontology	Ontology for surveys based on the Coney data model. @en
	ostop - Ordnance Survey Topography Ontology	http://www.ordnancesurvey.co.uk/ontology/Topography/v0.1/Topography.owl	The general purpose of this ontology is to provide a library of high level concepts that are used by the other modules within the whole OS Topographic ontology. The ontology also describes the relationships and instances common to more than one module. @en
	sensur - The Semantic Survey Ontology (sensur)	http://purl.org/SemSur/	SemSur, the Semantic Survey Ontology, is a core ontology for describing individual research problems, approaches, implementations and evaluations in a structured, comparable way. @en
	disco - DDI-RDF Discovery Vocabulary	http://rdf-vocabulary.ddialliance.org/discovery	A metadata vocabulary for documenting research and survey data @en
	g50k - 50K Gazetteer Vocabulary	http://data.ordnancesurvey.co.uk/ontology/50kGazetteer/	A vocabulary developed to describe the Ordnance Survey 50k Gazetteer linked data @en

On the right side of the search results, there are two filters:

- Type**
 - vocabulary (6)
 - property/class >
 - agent >
- Tag**
 - Methods (3)
 - Geography (2)
 - Geometry (1)
- Language**
 - English (3)
 - French (1)
 - German (1)

Figure 4.1: Search Results for “Survey” on Linked Open Vocabularies Website

4.2 Modelling Ontology for Survey Using Protégé

Figure 4.2 shows the student feedback form for which the new vocabulary is described. It comprises 9 *likert-scale* questions and 3 *open-ended* questions.

4.2.1 Classes for Survey Ontology

All the classes can be created and described using the steps described in Section 3.2.2.1. The resultant class hierarchy for the survey ontology would be as shown in Figure 4.3. A Survey can have Survey Collection, Survey Respondent, and Type of Survey which have been considered for this ontology as well.

The SurveyCollection class corresponds to a collection of survey responses that were collected for a particular course-offering (CourseOffering class). The type of survey used is feedback survey (FeedbackSurvey class, sub-class of Survey) with Dr. Hept-

Student Feedback Form

(adapted from <http://cft.vanderbilt.edu/teaching-guides/reflecting/student-feedback/#inclass>)

Course: _____ **Instructor:** Daryl Hepting **Date:** _____

1 = Never; 5 = Always

- | | | | | | | |
|---|---|---|---|---|---|---|
| 1 | The instructor is well-prepared for class. | 1 | 2 | 3 | 4 | 5 |
| 2 | The instructor clearly communicates his expectations for student preparation and participation. | 1 | 2 | 3 | 4 | 5 |
| 3 | The instructor uses class time effectively. | 1 | 2 | 3 | 4 | 5 |
| 4 | The instructor has clear expectations for assigned work. | 1 | 2 | 3 | 4 | 5 |
| 5 | The instructor encourages student participation. | 1 | 2 | 3 | 4 | 5 |
| 6 | The instructor clearly answers questions. | 1 | 2 | 3 | 4 | 5 |
| 7 | The instructor treats students with respect. | 1 | 2 | 3 | 4 | 5 |
| 8 | The instructor effectively directs and stimulates discussion. | 1 | 2 | 3 | 4 | 5 |
| 9 | The instructor effectively encourages students to ask questions and give answers. | 1 | 2 | 3 | 4 | 5 |

What do you like best about this course?

What would you like to change about this course?

What do you think the instructor's greatest strengths are?

Figure 4.2: The Student Feedback Form, from Hepting [27].

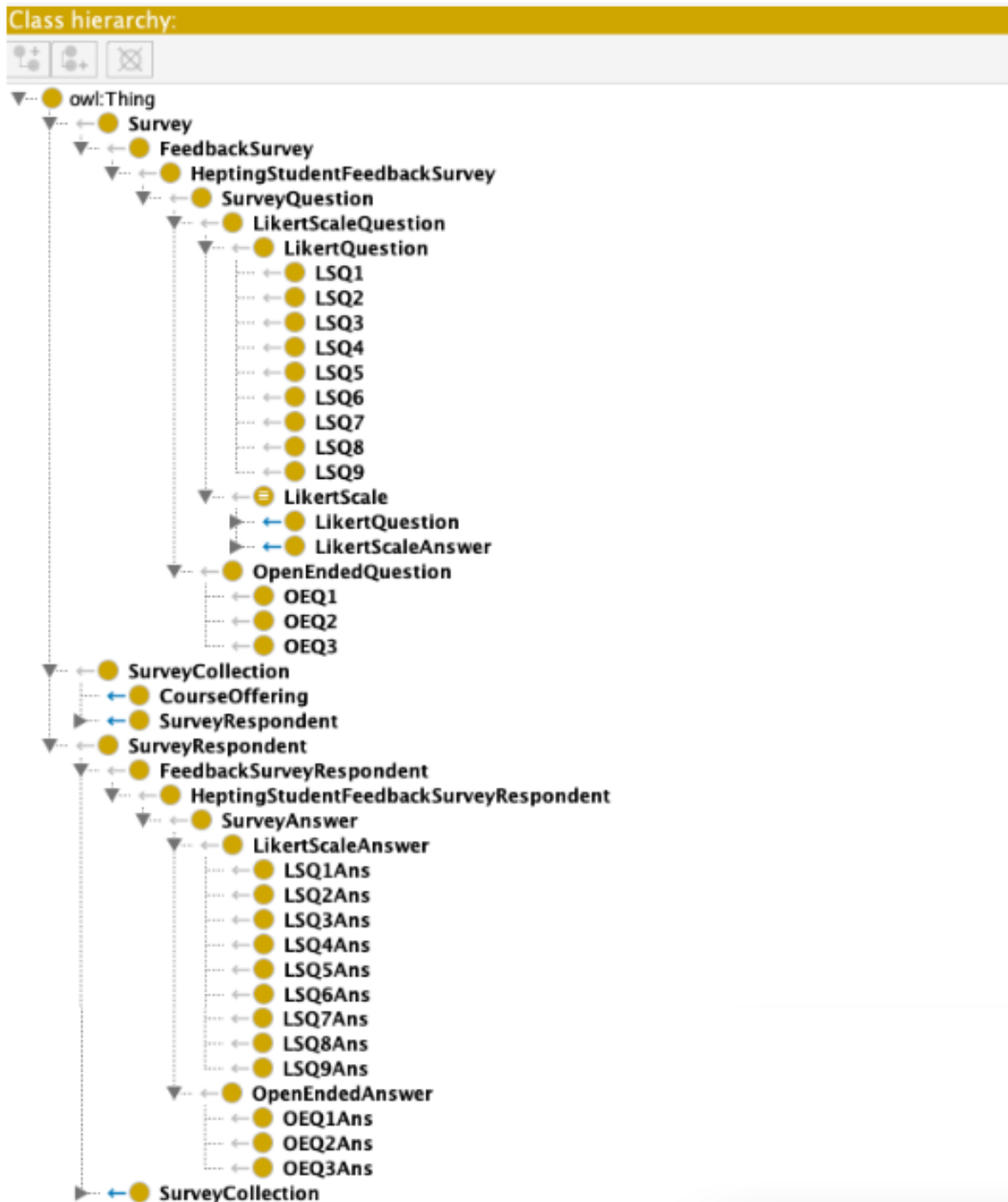


Figure 4.3: Class Hierarchy in Survey Ontology

ing’s feedback survey as the specific survey instrument (`HeptingStudentFeedbackSurvey` class) and responses to the survey (`HeptingStudentFeedbackSurveyRespondent` class).

The survey questions and survey answers have classes corresponding to various forms of question-answer. Generally, feedback surveys consist of two types of questions, namely, *Likert Questions* and *Open-ended Questions*. The type of questions might differ depending on the designer of the survey. The feedback survey in this research has Likert and Open-ended questions. The classes described for the question and answer are `SurveyQuestion` and `SurveyAnswer` as sub-classes of `HeptingStudentFeedbackSurvey` and `HeptingStudentFeedbackSurveyRespondent` respectively. For each survey question, there is an answer, where both of them are considered as distinct classes. The details of `SurveyQuestion` and `SurveyAnswer` classes are in following sections.

4.2.1.1 The “SurveyQuestion” Class

Figure 4.4 shows the `SurveyQuestion` Class from survey ontology. Each question in survey has one question label whose datatype is string. It has two sub-classes, one for each of the `OpenEndedQuestion` and `LikertScaleQuestion` classes.

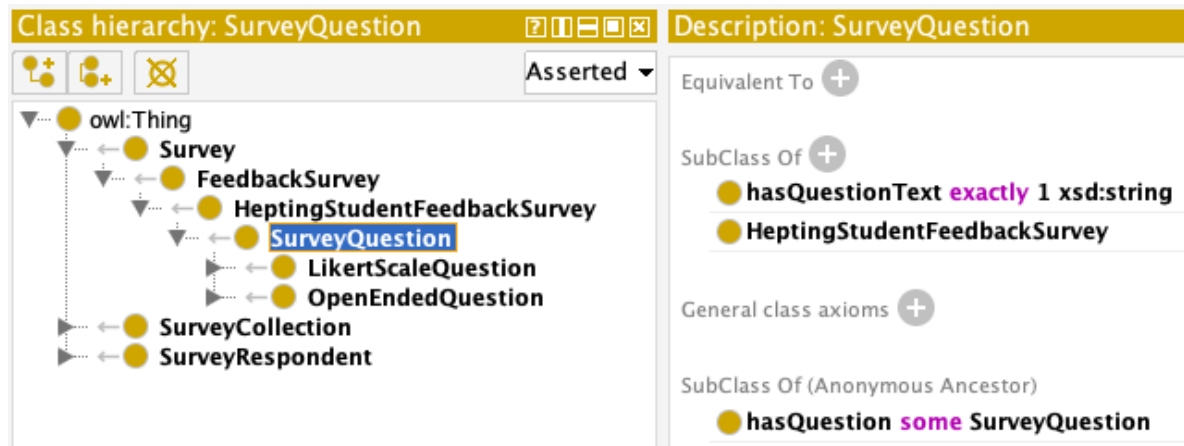


Figure 4.4: SurveyQuestion Class in Survey Ontology

The `OpenEndedQuestion` class describes the set of 3 open-ended questions on the

survey instrument where each question has it's own class (sub-class of `OpenEndedQuestion`), namely, `OEQ1`, `OEQ2`, and `OEQ3` with question labels, *What do you like best about the course?*, *What would you like to change about the course?*, and *What do you think the instructor's greatest strengths are?* respectively.

Likert-scale questions are the questions for which the answers are from the preset list of answers. The `LikertScaleQuestion` class represents a set of 9 likert-scale questions from the survey instrument. Each likert-scale question has a `LikertQuestion` and a `LikertScale`. where `LikertQuestion` has the 9 questions each as a distinct class. `LSQ1`, `LSQ2`, . . . , and `LSQ9` are the classes with question label representing the question.

The `LikertScale` class describes the Likert scale used by Hepting in his survey. The Likert scale has degrees of agreement-either in the form of label or a number, which determine the type of likert-scale used. Here, there are 5 options, 1, 2, 3, 4, and 5 where 1 = Never and 5 = Always (see Figure 4.2). In the vocabulary, each of these degrees of agreement has been described as an individual of the `LikertScale` class as shown in Figure 4.5.

4.2.1.2 The “SurveyAnswer” Class

The `SurveyAnswer` class represents the responses to the questions of the survey. It has two sub-classes, one for each of the `OpenEndedAnswer` and `LikertScaleAnswer` classes. The `OpenEndedAnswer` class represents the set of answers for 3 open-ended questions. Open-ended questions have unpredictable answers. Answer to each question is described by a distinct class (sub-class of `OpenEndedAnswer`) that consists of a string, namely, `OEQ1Ans`, `OEQ2Ans`, and `OEQ3Ans`.

The `LikertScaleAnswer` class represents the set of answers for 9 likert-scale questions. Answer to each question would be a value belonging to the `LikertScale`, which is between 1 and 5. In case of no answer, the data field will be left blank. There are

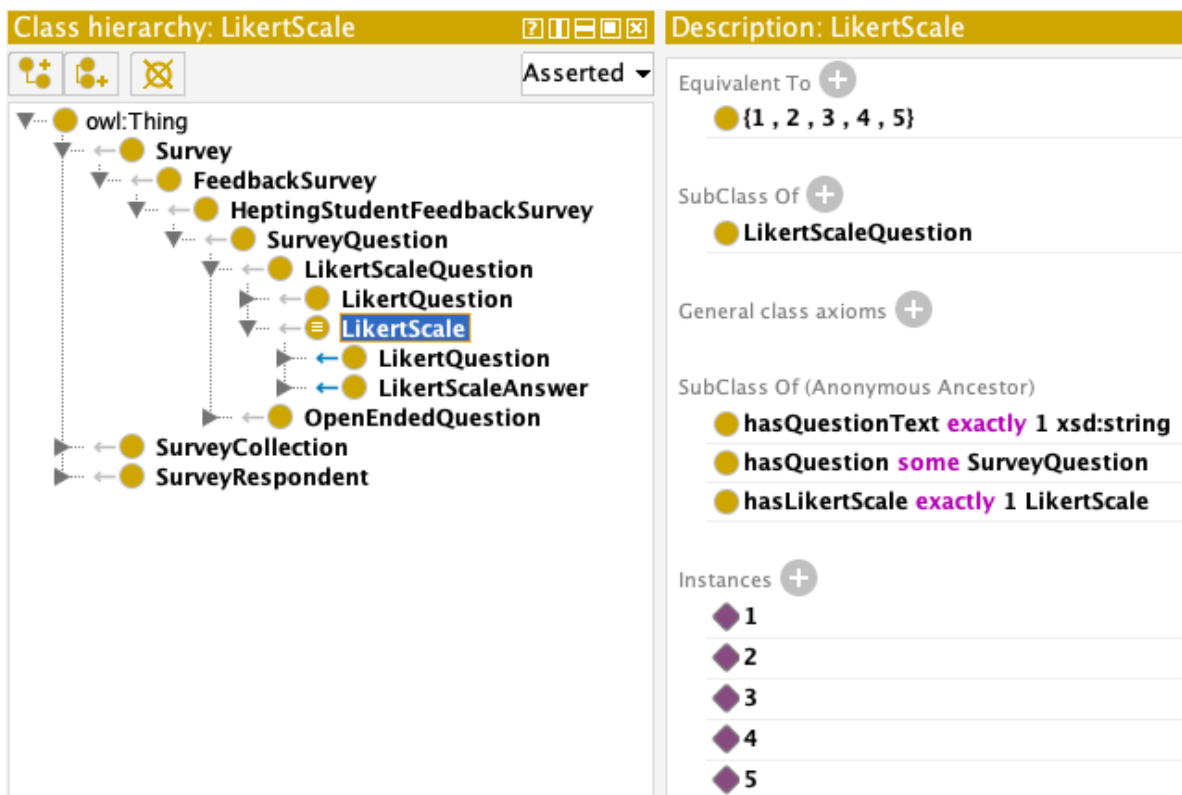


Figure 4.5: LikertScale Class in Survey Ontology

9 sub-classes of the `LikertScaleAnswer` class, namely, `LSQ1Ans`, `LSQ2Ans`, . . . , and `LSQ9Ans`. The next step is to describe properties.

4.2.2 Object Properties for Survey Ontology

Using the method of naming and building object properties described in Section 3.2.2.2, some of the primary object properties are described for the survey ontology which are listed in Table 4.1 along with their domain and range classes. The parent node for all object properties is `owl:TopObjectProperty`.

Object Property	Domain	Range
<code>hasCollection</code>	<code>CourseOffering</code>	<code>SurveyCollection</code>
<code>hasCourseOffering</code>	<code>HeptingStudentFeedbackSurvey</code>	<code>CourseOffering</code>
<code>hasRespondent</code>	<code>SurveyCollection</code>	<code>SurveyRespondent</code>
<code>isTypeOf</code>	<code>FeedbackSurvey</code>	<code>Survey</code>
<code>hasQuestion</code>	<code>Survey</code>	<code>SurveyQuestion</code>
<code>hasLikertScale</code>	<code>LikertQuestion</code> , <code>LikertScaleAnswer</code>	<code>LikertScale</code>
<code>hasEligibleRespondents</code>	<code>SurveyCollection</code>	

Table 4.1: Object Properties and Their Domain and Range

4.2.3 Data Properties for Survey Ontology

Data properties for survey ontology have been described following the method described in Section 3.2.2.3 which are listed in Table 4.2 along with their domain and range. The parent node of all the data properties is `owl:topDataProperty`. The properties for Likert answers restrict the datatype to positive integer and to string for open-ended answers.

4.2.4 Individuals

While dealing with ontologies, an instance of class is termed as an **Individual**. In protégé, an individual can be described as an instance of class using the Description

Data Property	Domain	Range
hasLSQ1Ans	LSQ1	xsd:positiveInteger
hasLSQ2Ans	LSQ2	xsd:positiveInteger
hasLSQ3Ans	LSQ3	xsd:positiveInteger
hasLSQ4Ans	LSQ4	xsd:positiveInteger
hasLSQ5Ans	LSQ5	xsd:positiveInteger
hasLSQ6Ans	LSQ6	xsd:positiveInteger
hasLSQ7Ans	LSQ7	xsd:positiveInteger
hasLSQ8Ans	LSQ8	xsd:positiveInteger
hasLSQ9Ans	LSQ9	xsd:positiveInteger
hasOEQ1Ans	OEQ1	xsd:string
hasOEQ2Ans	OEQ2	xsd:string
hasOEQ3Ans	OEQ3	xsd:string
hasQuestionText	SurveyQuestion	xsd:string

Table 4.2: Data Properties and Their Domain and Range

view of the Class tab as shown in Figure 4.5 for **LikertScale** class. A new individual can be described using the Individuals view of the Entities tab where as the property assertions view can be used to describe the property restrictions. In this research, the individuals have been described as a part of data file (Appendix B).

The final class hierarchy (see Figure 4.3) can be obtained for feedback survey in protégé using the HermiT 1.4.3.456 reasoner. Figure 4.6 shows blue back arrows in front of the yellow solid dot, which indicates class, to denote relationship among classes other than that of SubClassOf. This feature can be enabled and disabled by selecting **Display relationship in class hierarchy** option from the View menu of protégé.

4.2.5 Results of FAIR Evaluator

After describing, the vocabulary was tested against the FAIR principles to evaluate its usability using the “FOOPS! Ontology Pitfall Scanner for FAIR (Beta)”. The results in Figure 4.7 show that 30% of the **survey.owl** vocabulary obeys the FAIR principles [42].

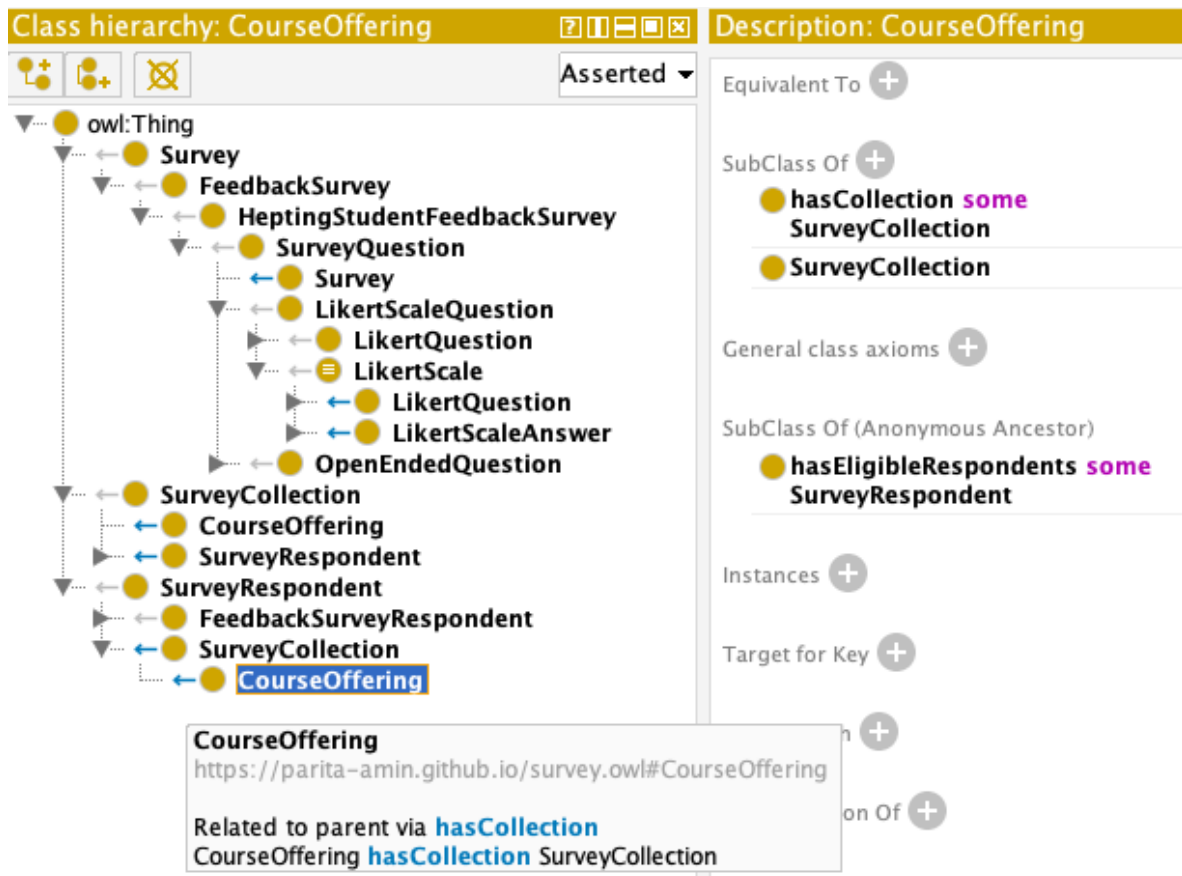


Figure 4.6: Relationship among classes in Survey Ontology

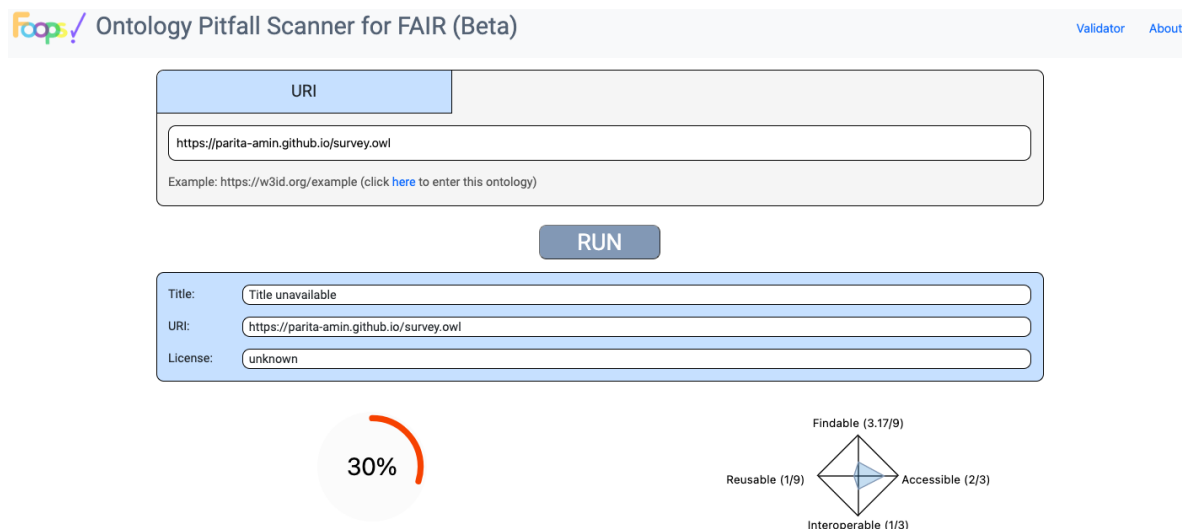


Figure 4.7: The Results of FAIR Evaluator for Survey Ontology

The results in Figure 4.7 depicts that the `survey.owl` vocabulary is 35% Findable, 66% Accessible, 33% Interoperable, and 11% Reusable.

4.3 Data Validation Using SHACL

SHACL4P is a protégé plugin for SHACL, which uses *shapes* to describe constraints. SHACL can be used to validate the manually entered data which are stored in spreadsheets. For feedback survey, the likert answers can only have values 1, 2, 3, 4, 5, or No Answer (blank cell). To restrict the values of data properties corresponding to likert questions in Table 4.2, SHACL4P has been used.

Protégé offers a view for SHACL editor as well as SHACL constraint violations. For successful working of SHACL4P, a reasoner should be enabled. The SHACL editor provides three buttons, namely, **Open**, **Save**, and **Validate**. After opening the shapes file (**Open** button), changes can be made and saved (**Save** button). Where as the **Validate** button allows users to validate the data. Figure 4.8 shows a part of `SurveyShapes.txt` file opened in SHACL editor of protégé for the data property `hasLSQ1Ans`.

On clicking the **Validate** button, the summary of violated constraints is displayed on the SHACL constraint violations view in protégé. It contains the details like **Severity**, **SourceShape**, **Message**, **FocusNode**, **Path**, and **Value** as shown in Figure 4.9.

4.4 Querying Survey Ontology in Protégé

The ontology vocabularies can be queried using query languages like DL query² and SPARQL. Protégé offers a plugin for DL as well as SPARQL queries which are discussed in this section.

²https://protegewiki.stanford.edu/wiki/DL_Query

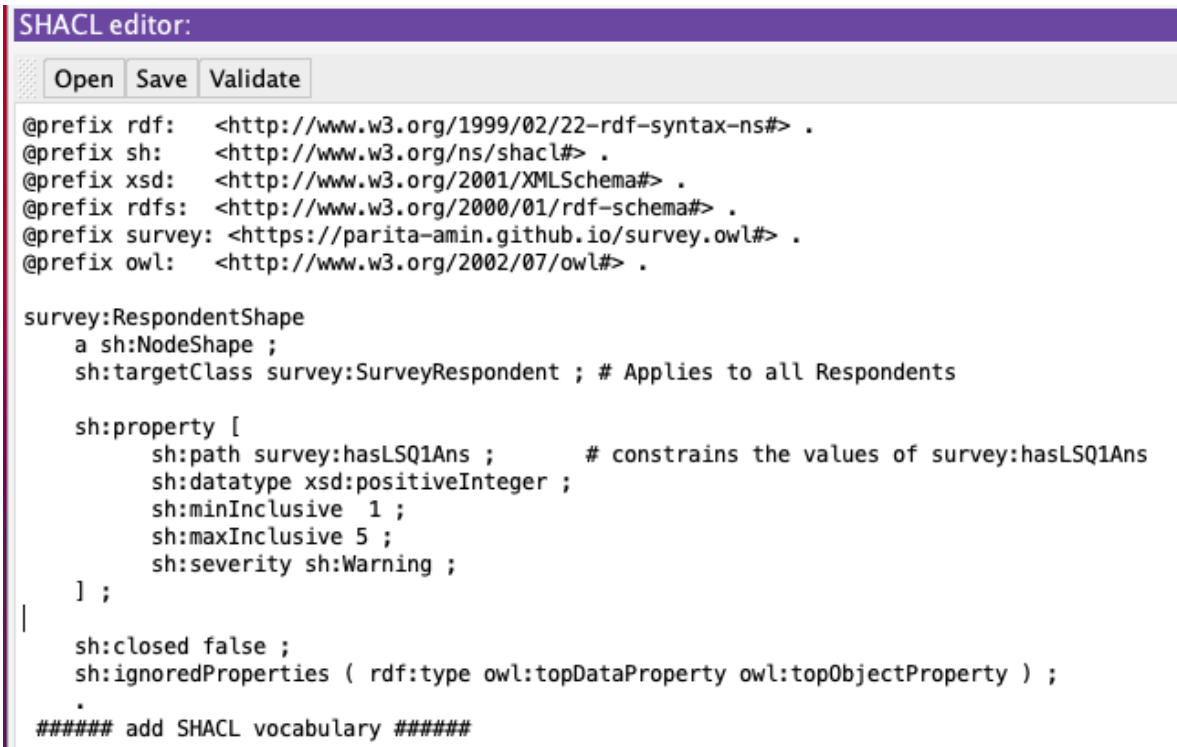


Figure 4.8: Data Validation using SHACL Editor

SHACL constraint violations: 27					
Severity	SourceShape	Message	FocusNode	Path	Value
http://www.w3.org/ns/shacl#Warning	ef650c66...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ1Ans	-999
http://www.w3.org/ns/shacl#Warning	ef650c66...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ1Ans	-999
http://www.w3.org/ns/shacl#Warning	ef650c66...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ1Ans	-999
http://www.w3.org/ns/shacl#Warning	41609a0...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ2Ans	-999
http://www.w3.org/ns/shacl#Warning	41609a0...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ2Ans	-999
http://www.w3.org/ns/shacl#Warning	41609a0...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ2Ans	-999
http://www.w3.org/ns/shacl#Warning	bae1b61c...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ3Ans	-999
http://www.w3.org/ns/shacl#Warning	bae1b61c...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ3Ans	-999
http://www.w3.org/ns/shacl#Warning	bae1b61c...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ3Ans	-999
http://www.w3.org/ns/shacl#Warning	364016b...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ4Ans	-999
http://www.w3.org/ns/shacl#Warning	364016b...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ4Ans	-999
http://www.w3.org/ns/shacl#Warning	364016b...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ4Ans	-999
http://www.w3.org/ns/shacl#Warning	14e4fddc...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ5Ans	-999
http://www.w3.org/ns/shacl#Warning	14e4fddc...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ5Ans	-999
http://www.w3.org/ns/shacl#Warning	14e4fddc...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ5Ans	-999
http://www.w3.org/ns/shacl#Warning	7eaf5ab0...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ6Ans	-999
http://www.w3.org/ns/shacl#Warning	7eaf5ab0...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ6Ans	-999
http://www.w3.org/ns/shacl#Warning	7eaf5ab0...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ6Ans	-999
http://www.w3.org/ns/shacl#Warning	18601df4...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ7Ans	-999
http://www.w3.org/ns/shacl#Warning	18601df4...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ7Ans	-999
http://www.w3.org/ns/shacl#Warning	18601df4...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ7Ans	-999
http://www.w3.org/ns/shacl#Warning	69a4641...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ8Ans	-999
http://www.w3.org/ns/shacl#Warning	69a4641...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ8Ans	-999
http://www.w3.org/ns/shacl#Warning	69a4641...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ8Ans	-999
http://www.w3.org/ns/shacl#Warning	8764840...	Cannot compare with 1	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ9Ans	-999
http://www.w3.org/ns/shacl#Warning	8764840...	Cannot compare with 5	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ9Ans	-999
http://www.w3.org/ns/shacl#Warning	8764840...	Value must be a valid literal of type positiveInteger	https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R55	https://parita-amin.github.io/survey.owl#hasLSQ9Ans	-999

Figure 4.9: SHACL Constraint Violations

4.4.1 DL Query

Vasilevsky³ defines DL query as a class expression that is constructed using constructs such as **and**, **or** (corresponding to set intersection) and **some**. Users can rapidly check that class definitions encompass the right subclasses by running them via the DL Query tab.

Alternately, one can examine arbitrary descriptions for class membership without having to build placeholders for specified classes. The DL Query tab can be opened using Tabs option under Windows menu. It is important to configure a reasoner to successfully express a DL query. Figure 4.10 shows expression of a sample DL query using the survey ontology in protégé.

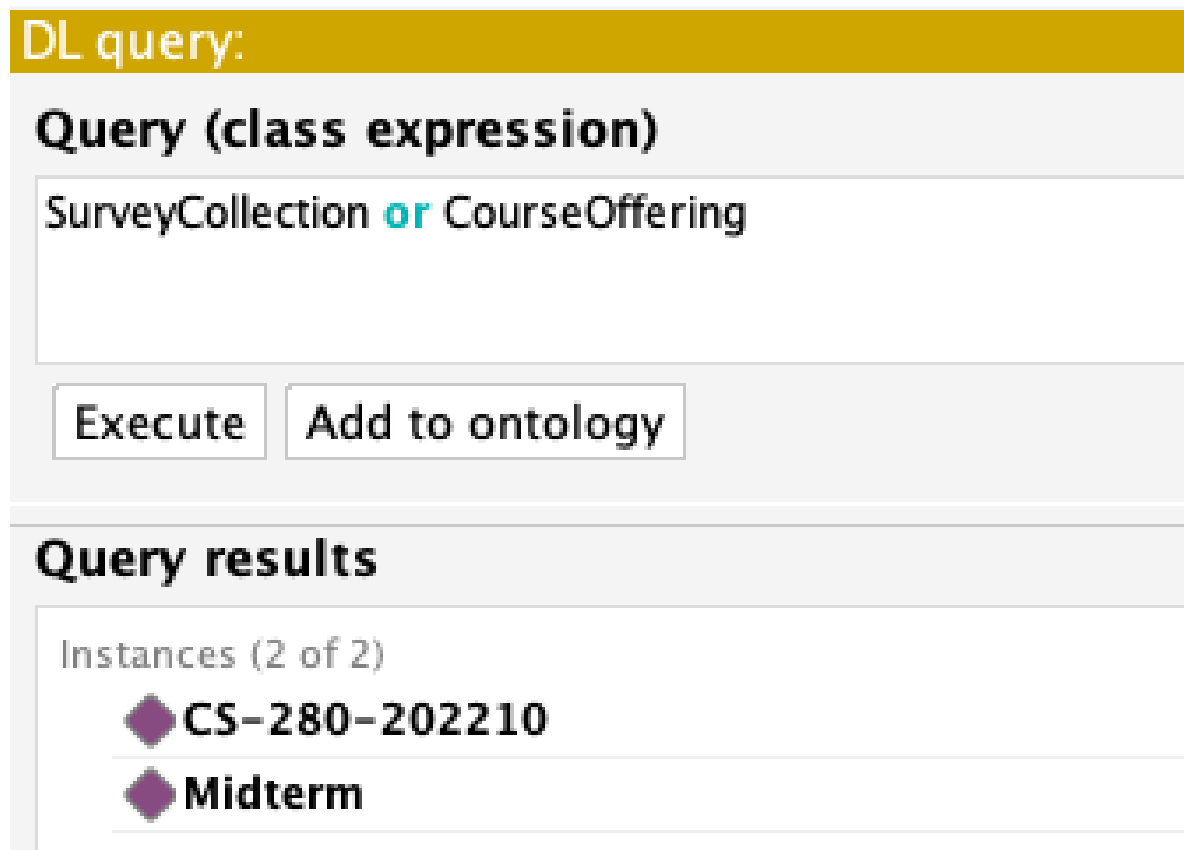


Figure 4.10: DL Query on Survey Ontology

³https://ontology101tutorial.readthedocs.io/en/latest/EXERCISE_BasicDL_Queries.html

4.4.2 SPARQL Query

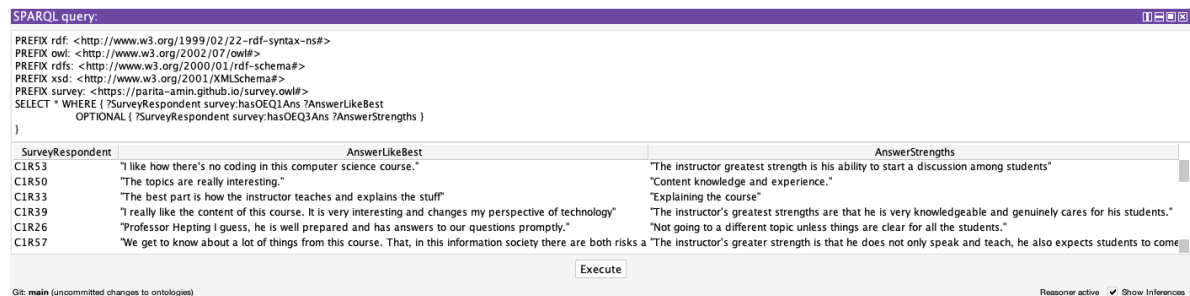
According to W3C [21], SPARQL queries use conjunctions, disjunctions, aggregation, subqueries, negation, and other operations to define queries and optional graph patterns. Protégé offers plugins for SPARQL Query and Snap SPARQL Query.

4.4.2.1 Using SPARQL Query Plugin

SPARQL plugin in protégé is used to express simple queries on ontologies like displaying classes and subclasses. For instance, consider the following query:

```
SELECT * WHERE { ?SurveyRespondent survey:hasOEQ1Ans
                  ?AnswerLikeBest
                  OPTIONAL { ?SurveyRespondent survey:hasOEQ3Ans
                              ?AnswerStrengths }
}
```

Figure 4.11 shows the result of expression of SPARQL query on survey ontology. Here, the query is used to display `SurveyRespondent`, `hasOEQ1Ans`, and `hasOEQ3Ans` with column titles `SurveyRespondent`, `AnswerLikeBest` and `AnswerStrengths` respectively.



The screenshot shows the SPARQL Query interface in Protégé. The query is as follows:

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX survey: <https://parita-amin.github.io/survey.owl#>
SELECT * WHERE { ?SurveyRespondent survey:hasOEQ1Ans ?AnswerLikeBest
                  OPTIONAL { ?SurveyRespondent survey:hasOEQ3Ans ?AnswerStrengths }
}
```

The results are displayed in a table with three columns: `SurveyRespondent`, `AnswerLikeBest`, and `AnswerStrengths`.

SurveyRespondent	AnswerLikeBest	AnswerStrengths
C1R53	"I like how there's no coding in this computer science course."	"The instructor's greatest strength is his ability to start a discussion among students"
C1R50	"The topics are really interesting."	"Content knowledge and experience."
C1R33	"The best part is how the instructor teaches and explains the stuff"	"Explaining the course"
C1R39	"I really like the content of this course. It is very interesting and changes my perspective of technology"	"The instructor's greatest strengths are that he is very knowledgeable and genuinely cares for his students."
C1R26	"Professor Hepting I guess, he is well prepared and has answers to our questions promptly."	"Not going to a different topic unless things are clear for all the students."
C1R57	"We get to know about a lot of things from this course. That, in this information society there are both risks a	"The instructor's greater strength is that he does not only speak and teach, he also expects students to come

At the bottom of the interface, there is an "Execute" button and a status bar showing "Git: main (uncommitted changes to ontologies)" and "Reasoner active Show Inferences".

Figure 4.11: SPARQL Query on Survey Ontology

4.4.2.2 Using Snap SPARQL Query Plugin

The Snap SPARQL plugin in protégé is generally used to express comparatively complex queries on ontologies like displaying description of individuals. For instance, consider the following query:

```
SELECT * WHERE { ?Respondent rdf:type survey:SurveyRespondent .  
    OPTIONAL {?Respondent survey:hasLSQ6Ans  
        ?The_instructor_clearly_answers_questions}  
    } ORDER BY ?SurveyRespondent
```

Figure 4.12 shows the result of expression of Snap SPARQL query on survey ontology. It is necessary to enable a reasoner for successful working of Snap SPARQL query. Here, the query is used to display **SurveyRespondent** and **hasLSQ6Ans** with column titles **?Respondent** and **?The_instructor_clearly_answers_questions** respectively.

4.5 Mapping Data from Spreadsheets to RDF For Survey using OpenRefine

Primarily, OpenRefine⁴ has been used to convert the data stored in spreadsheets to Turtle and RDF files using the methodology described in Section 3.3. Using survey ontology in OpenRefine, the CSV file consisting of survey responses is converted into Turtle and RDF/XML files.

After importing the CSV and creating a new project, the next step is to manage RDF Schema alignment to describe the data stored in CSV in terms of linked data vocabulary. On selecting **Edit RDF Skeleton** from the menu of RDF Extension, an

⁴<https://openrefine.org/>

Snap SPARQL Query:

```

PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX survey: <https://parita-amin.github.io/survey.owl#>

SELECT * WHERE {
  ?Respondent rdf:type survey:SurveyRespondent .
  OPTIONAL {
    ?Respondent survey:hasLSQ6Ans
    ?The_instructor_clearly_answers_questions
  }
}
ORDER BY ?SurveyRespondent

```

Execute

?Respondent	?The_instructor_clearly_answers_questions
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R33>	5
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R32>	5
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R35>	5
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R34>	4
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R37>	2
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R36>	4
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R39>	4
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R38>	5
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R31>	5
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R30>	4
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R22>	5
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R21>	4
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R24>	5
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R23>	4
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R26>	5
<https://parita-amin.github.io/survey-data/CS-280-202210.ttl#C1R25>	4

59 results

Figure 4.12: Snap SPARQL Query on Survey Ontology

RDF Schema alignment dialog box appears which offers two options: **RDF Skeleton** and **RDF Preview**. The base URI for the RDF schema alignment has been set to:

Base URI: `https://parita-amin.github.io/` (4.1)

4.5.1 RDF Skeleton

Figures 4.13, 4.14, and 4.15 show RDF skeleton for CSV using survey ontology. It consists of a list of prefixes along with options for adding and managing prefixes; URI for index count; and the fields in the project with an option of using linked data vocabularies to describe them.

RDF Schema alignment

The RDF schema alignment skeleton below specifies how the RDF data that will get generated from your grid-shaped data. The cells in each record of your data will get placed into nodes within the skeleton. Configure the skeleton by specifying which column to substitute into which node.

Base URI: `https://parita-amin.github.io/` [Edit](#)

RDF skeleton

RDF Preview

Available prefixes:

rdf owl rdfs foaf survey xsd

+ Add

⚙ Manage

Username URI

✕

survey:FeedbackSurvey

Add type

✕

survey:hasCourseOffering

→

Add property

□

Course URI

⊕...

Add type

Course URI

✕

survey:CourseOffering

Add type

✕

survey:hasCollection

→

Add property

□

Group URI

⊕...

Add type

Group URI

✕

survey:SurveyCollection

Add type

✕

survey:hasEligibleRespondents

→

✕

survey:hasRespondent

→

Add property

□

100

□

Response

URI

⊕...

Add type

Figure 4.13: RDF Skeleton for CSV using Survey Ontology, 1 of 3

57

RDF skeleton

RDF Preview

Available prefixes:

rdf owl rdfs foaf survey xsd

+

Add

⚙

Manage

Response URI

✕

survey:SurveyRespondent

Add type

✕

➤

survey:hasLSQ1Ans

➔

✕

➤

survey:hasLSQ2Ans

➔

✕

➤

survey:hasLSQ3Ans

➔

✕

➤

survey:hasLSQ4Ans

➔

✕

➤

survey:hasLSQ5Ans

➔

☐

Q01_Rate instructor->1. The instructor is well-prepared for class Cell

☐

Q01_Rate instructor->2. The instructor clearly communicates his expectations for student preparation and participation Cell

☐

Q01_Rate instructor->3. The instructor uses class time effectively Cell

☐

Q01_Rate instructor->4. The instructor has clear expectations for assigned work Cell

☐

Q01_Rate instructor->5. The instructor encourages student participation Cell

Figure 4.14: RDF Skeleton for CSV using Survey Ontology, 2 of 3

✕

➤

survey:hasLSQ6Ans

➔

✕

➤

survey:hasLSQ7Ans

➔

✕

➤

survey:hasLSQ8Ans

➔

✕

➤

survey:hasLSQ9Ans

➔

✕

➤

survey:hasOEQ1Ans

➔

✕

➤

survey:hasOEQ2Ans

➔

✕

➤

survey:hasOEQ3Ans

➔

Add property

☐

Q01_Rate instructor->6. The instructor clearly answers questions Cell

☐

Q01_Rate instructor->7. The instructor treats students with respect Cell

☐

Q01_Rate instructor->8. The instructor effectively directs and stimulates discussion Cell

☐

Q01_Rate instructor->9. The instructor effectively encourages students to ask questions and give answers Cell

☐

Q02_Like best Cell

☐

Q03_Change Cell

☐

Q04_Strengths Cell

Add another root node

Save

OK

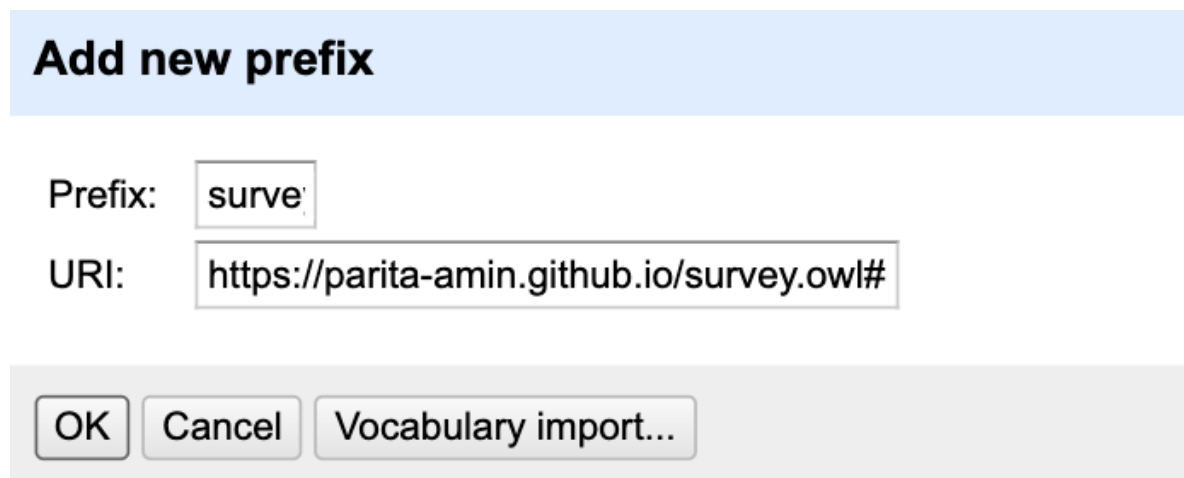
Cancel

Figure 4.15: RDF Skeleton for CSV using Survey Ontology, 3 of 3

4.5.1.1 Prefixes

OpenRefine offers some default prefixes, namely, `rdf`, `owl`, `rdfs`, and `foaf`, which can be used and managed as necessary. It also allows adding other prefixes which can be done by clicking on **+Add** option as shown in Figure 4.16. Following prefixes along with URIs have been added to the list:

```
survey : https://parita-amin.github.io/survey.owl#  
xsd : http://www.w3.org/2001/XMLSchema#
```



Add new prefix

Prefix:

URI:

Figure 4.16: Add New Prefix for Survey Ontology in OpenRefine

The prefixes can be deleted and/or refreshed using **Manage** option as shown in Figure 4.17.

4.5.1.2 RDF Node for Index

Index is a unique identifier for the number of records in the table. The settings of the RDF Node for index offer options for **Use content...**, **Content used...**, and **Use expression...**. On clicking **Preview edit**, format of expression to display records can be modified.

List of defined prefixes

Add prefix

Prefix	URI	Delete	Refresh
rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#	Delete	Refresh
owl	http://www.w3.org/2002/07/owl#	Delete	Refresh
survey	https://parita-amin.github.io/survey.owl#	Delete	Refresh
rdfs	http://www.w3.org/2000/01/rdf-schema#	Delete	Refresh
foaf	http://xmlns.com/foaf/0.1/	Delete	Refresh
xsd	http://www.w3.org/2001/XMLSchema#	Delete	Refresh
	https://parita-amin.github.io/	Delete	Refresh

OK

Cancel

Figure 4.17: Manage Prefixes in OpenRefine

4.5.1.3 Describing Fields in Terms of Linked Data

The next step is describing the CSV data in terms of linked open vocabularies. `Available prefixes` allows using vocabularies described in the list of ontologies. It offers a list of classes and properties along with the ontology prefixes that match the input string. The vocabulary can be selected for each field of the CSV to represent the data.

For instance, `survey:hasLSQ1Ans` property can be assigned to the data corresponding to the field, *Q01_Rate instructor->1. The instructor is well-prepared for class* of the CSV.

4.5.2 RDF Preview

Once the RDF skeleton is set, `RDF Preview` can be used to view the description of CSV data. The fields of data that were not described in RDF skeleton, do not appear in

RDF Preview. It consists of a Turtle representation of the first 10 rows of the resultant linked data.

Figures 4.18 and 4.19 depict an RDF Preview of using survey vocabulary to describe the data stored in CSV. It includes all the prefixes for vocabularies from **Available prefixes** and with their URIs along with the sample representation of the result of using RDF Skeleton to describe data.

RDF Schema alignment

The RDF schema alignment skeleton below specifies how the RDF data that will get generated from your grid-shaped data. The cells in each record of your data will get placed into nodes within the skeleton. Configure the skeleton by specifying which column to substitute into which node.

Base URI: <https://parita-amin.github.io/> [Edit](#)

RDF skeleton

RDF Preview

This is a sample `Turtle` representation of (up-to) the *first 10* rows

```

@prefix : <https://parita-amin.github.io/> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix survey: <https://parita-amin.github.io/survey.owl#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .

:HeptingStudentFeedbackSurvey a survey:FeedbackSurvey;
  survey:hasCourseOffering :CS-280-202210 .

:CS-280-202210 a survey:CourseOffering;
  survey:hasCollection :Midterm .

:Midterm a survey:SurveyCollection;
  survey:hasEligibleRespondents "100"^^xsd:int;
  survey:hasRespondent :C1R1, :C1R2, :C1R3, :C1R4, :C1R5, :C1R6 .

```

Figure 4.18: RDF Preview for CSV using Survey Ontology, 1 of 2

Any modifications in RDF skeleton will reflect on the RDF preview. Once the data in CSV has been converted to linked data, it can be stored as RDF/XML and Turtle.

4.6 Publishing Survey Vocabulary with License

As mentioned in Section 3.4, it is necessary to specify an appropriate license for linked open data. When there is a clear declaration regarding the source, ownership, and conditions associated to the use of the public data, data reuse is more likely to

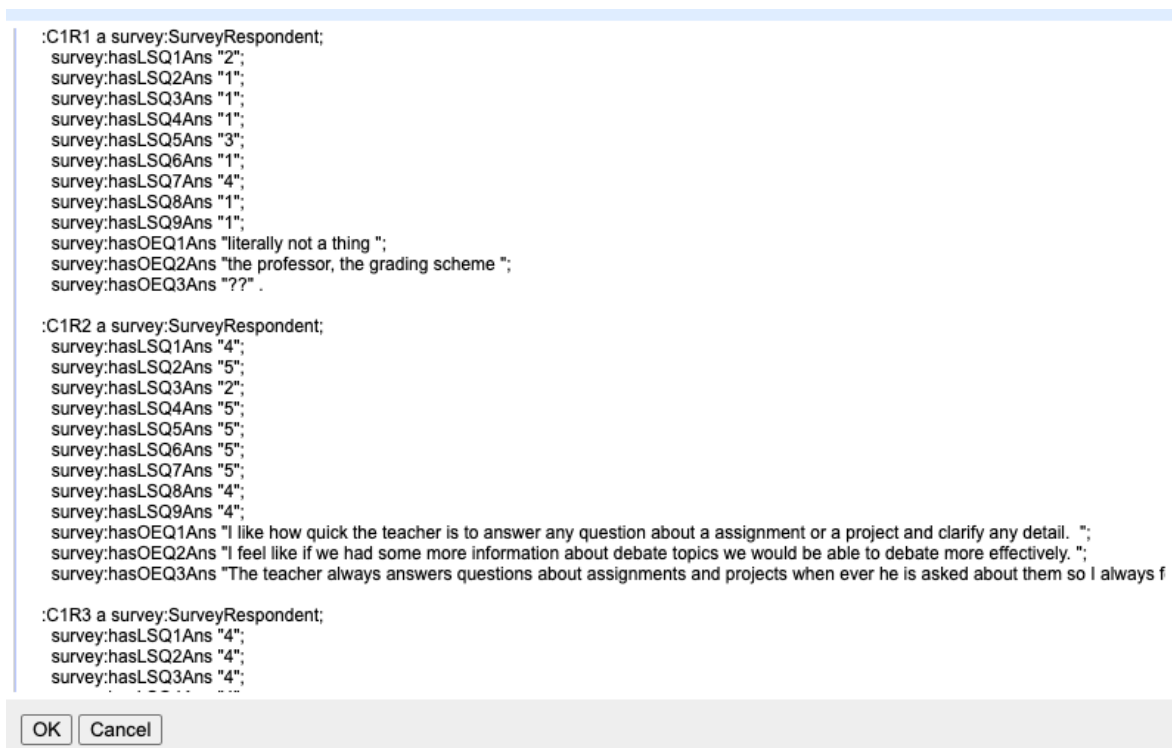


Figure 4.19: RDF Preview for CSV using Survey Ontology, 2 of 2

happen.

- Publishing Survey Vocabulary:

The linked open vocabulary for feedback survey that Dr. Hepting solicits from his students in the classes he teaches, can be accessed as Turtle file from my website⁵ or can be viewed by visiting the link to the Turtle file on my website⁶ (see Figure 4.20).

- License:

The linked open dataset discussed in this research has an **Attribution-ShareAlike 4.0 International**⁷ open license. The degree of transparency increases as more

⁵<https://parita-amin.github.io/sample.ttl>

⁶<https://parita-amin.github.io/evaluationofteaching.html>

⁷<https://creativecommons.org/licenses/by-sa/4.0/>

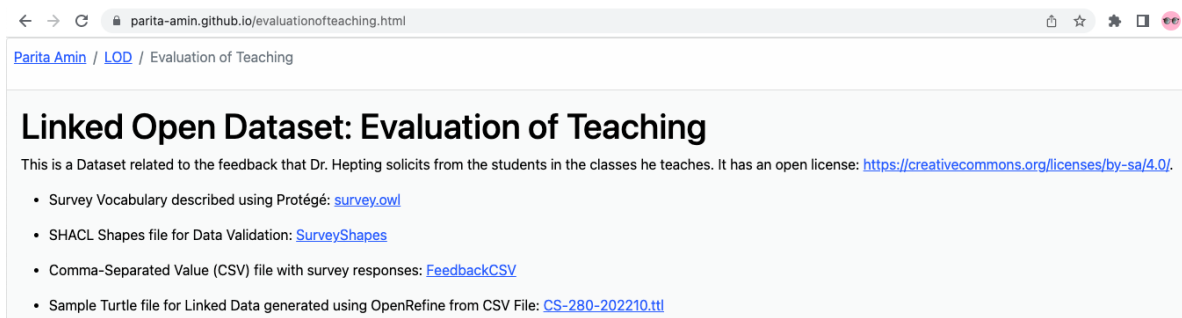


Figure 4.20: Link to Turtle File containing Survey Vocabulary

data become available under an open license. This makes it possible for the user to investigate the relationships in the data and use it effectively.

Chapter 5

Conclusion and Future Work

This Chapter presents a summary of the research undertaken and contained in this document. It identifies the contributions made in this thesis and opportunities for future work.

5.1 Conclusion

After motivating the topic in Chapter 1, Chapter 2 provided a detailed background study on the fundamental topics supporting LOD and the tools which facilitate its publication on the web. These included Semantic Web, Linked Open Data, RDF, and OWL ontologies. A methodology of transforming data contained in a spreadsheet into LOD available on the web of data was described, both in general terms (Chapter 3) and in the specific case of Hepting's student feedback survey (Chapter 4). The thesis can be considered as a guide for researchers who are interested in transforming their own data into LOD. In response to the question asked about whether this next web of data sufficiently attractive to use, I have found sufficiency with opportunities for improvement.

5.2 Future Work

While the vocabulary developed for Hepting’s survey is a first step in describing Likert items using Likert scales, this vocabulary could be improved by reusing parts of other existing vocabularies. Adding expressiveness to the vocabulary by modelling a range of Likert [40] scales with different labels and values would also be valuable.

OpenRefine was used in this thesis for mapping data to RDF. Although it has a solid user interface, RMLMapper may be preferred in future because it permits the mappings to be scripted and executed from a command line interface. Such an approach may be preferable for the same mapping applied to several spreadsheets.

Hepting has a substantial collection of feedback data from these surveys spanning several years. Conversion of all this data into LOD, in a way that permits effective querying and analysis, would be an important achievement. The latest survey collections have once again been done on paper, and it would be helpful to have a means to support data entry.

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Appendix A

OWL Vocabulary for Survey Data: survey.owl

OWL Vocabulary for Survey Data. This is the OWL file that describes the feedback survey data received.

```
1 @prefix : <https://parita-amin.github.io/survey.owl> .
2 @prefix owl: <http://www.w3.org/2002/07/owl#> .
3 @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
4 @prefix xml: <http://www.w3.org/XML/1998/namespace> .
5 @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
6 @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
7 @prefix survey: <https://parita-amin.github.io/survey.owl#> .
8 @base <https://parita-amin.github.io/survey.owl> .
9
10 <https://parita-amin.github.io/survey.owl> rdf:type owl:Ontology ;
11                                     owl:versionIRI
12                                     <https://parita-amin.github.io/survey.owl/1.0> .
13 #####
14 #      Object Properties
15 #####
16
17 ### https://parita-amin.github.io/survey.owl#hasCollection
18 survey:hasCollection rdf:type owl:ObjectProperty ;
19                     rdfs:domain survey:CourseOffering ;
20                     rdfs:range survey:SurveyCollection .
21
22
23 ### https://parita-amin.github.io/survey.owl#hasCourseOffering
24 survey:hasCourseOffering rdf:type owl:ObjectProperty ;
25                     rdfs:domain survey:HeptingStudentFeedbackSurvey
26                     ;
27                     rdfs:range survey:CourseOffering .
28
29 ### https://parita-amin.github.io/survey.owl#hasEligibleRespondents
```

```

30 survey:hasEligibleRespondents rdf:type owl:ObjectProperty ;
31                                rdfs:domain survey:SurveyCollection .
32
33
34 #### https://parita-amin.github.io/survey.owl#hasLikertScale
35 survey:hasLikertScale rdf:type owl:ObjectProperty ;
36                        rdfs:domain survey:LikertQuestion ,
37                                survey:LikertScaleAnswer ;
38                        rdfs:range survey:LikertScale .
39
40
41 #### https://parita-amin.github.io/survey.owl#hasQuestion
42 survey:hasQuestion rdf:type owl:ObjectProperty ;
43                    rdfs:domain survey:Survey ;
44                    rdfs:range survey:SurveyQuestion .
45
46
47 #### https://parita-amin.github.io/survey.owl#hasRespondent
48 survey:hasRespondent rdf:type owl:ObjectProperty ;
49                       rdfs:domain survey:SurveyCollection ;
50                       rdfs:range survey:SurveyRespondent .
51
52
53 #### https://parita-amin.github.io/survey.owl#isTypeOf
54 survey:isTypeOf rdf:type owl:ObjectProperty ;
55                 rdfs:domain survey:FeedbackSurvey ;
56                 rdfs:range survey:Survey .
57
58
59 #####
60 # Data properties
61 #####
62
63 #### https://parita-amin.github.io/survey.owl#hasLSQ1Ans
64 survey:hasLSQ1Ans rdf:type owl:DatatypeProperty ;
65                  rdfs:subPropertyOf owl:topDataProperty ;
66                  rdfs:domain survey:LSQ1 ;
67                  rdfs:range xsd:positiveInteger .
68
69
70 #### https://parita-amin.github.io/survey.owl#hasLSQ2Ans
71 survey:hasLSQ2Ans rdf:type owl:DatatypeProperty ;
72                  rdfs:subPropertyOf owl:topDataProperty ;
73                  rdfs:domain survey:LSQ2 ;
74                  rdfs:range xsd:positiveInteger .
75
76
77 #### https://parita-amin.github.io/survey.owl#hasLSQ3Ans
78 survey:hasLSQ3Ans rdf:type owl:DatatypeProperty ;
79                  rdfs:subPropertyOf owl:topDataProperty ;
80                  rdfs:domain survey:LSQ3 ;
81                  rdfs:range xsd:positiveInteger .

```

```

82
83
84 #### https://parita-amin.github.io/survey.owl#hasLSQ4Ans
85 survey:hasLSQ4Ans rdf:type owl:DatatypeProperty ;
86                    rdfs:subPropertyOf owl:topDataProperty ;
87                    rdfs:domain survey:LSQ4 ;
88                    rdfs:range xsd:positiveInteger .
89
90
91 #### https://parita-amin.github.io/survey.owl#hasLSQ5Ans
92 survey:hasLSQ5Ans rdf:type owl:DatatypeProperty ;
93                    rdfs:subPropertyOf owl:topDataProperty ;
94                    rdfs:domain survey:LSQ5 ;
95                    rdfs:range xsd:positiveInteger .
96
97
98 #### https://parita-amin.github.io/survey.owl#hasLSQ6Ans
99 survey:hasLSQ6Ans rdf:type owl:DatatypeProperty ;
100                   rdfs:subPropertyOf owl:topDataProperty ;
101                   rdfs:domain survey:LSQ6 ;
102                   rdfs:range xsd:positiveInteger .
103
104
105 #### https://parita-amin.github.io/survey.owl#hasLSQ7Ans
106 survey:hasLSQ7Ans rdf:type owl:DatatypeProperty ;
107                   rdfs:subPropertyOf owl:topDataProperty ;
108                   rdfs:domain survey:LSQ7 ;
109                   rdfs:range xsd:positiveInteger .
110
111
112 #### https://parita-amin.github.io/survey.owl#hasLSQ8Ans
113 survey:hasLSQ8Ans rdf:type owl:DatatypeProperty ;
114                   rdfs:subPropertyOf owl:topDataProperty ;
115                   rdfs:domain survey:LSQ8 ;
116                   rdfs:range xsd:positiveInteger .
117
118
119 #### https://parita-amin.github.io/survey.owl#hasLSQ9Ans
120 survey:hasLSQ9Ans rdf:type owl:DatatypeProperty ;
121                   rdfs:subPropertyOf owl:topDataProperty ;
122                   rdfs:domain survey:LSQ9 ;
123                   rdfs:range xsd:positiveInteger .
124
125
126 #### https://parita-amin.github.io/survey.owl#hasOEQ1Ans
127 survey:hasOEQ1Ans rdf:type owl:DatatypeProperty ;
128                   rdfs:subPropertyOf owl:topDataProperty ;
129                   rdfs:domain survey:OEQ1 ;
130                   rdfs:range xsd:string .
131
132
133 #### https://parita-amin.github.io/survey.owl#hasOEQ2Ans

```

```

134 survey:hasOEQ2Ans rdf:type owl:DatatypeProperty ;
135                      rdfs:subPropertyOf owl:topDataProperty ;
136                      rdfs:domain survey:OEQ2 ;
137                      rdfs:range xsd:string .
138
139
140 #### https://parita-amin.github.io/survey.owl#hasOEQ3Ans
141 survey:hasOEQ3Ans rdf:type owl:DatatypeProperty ;
142                      rdfs:subPropertyOf owl:topDataProperty ;
143                      rdfs:domain survey:OEQ3 ;
144                      rdfs:range xsd:string .
145
146
147 #### https://parita-amin.github.io/survey.owl#hasQuestionText
148 survey:hasQuestionText rdf:type owl:DatatypeProperty ;
149                      rdfs:domain survey:SurveyQuestion ;
150                      rdfs:range xsd:string .
151
152
153 #####
154 #      Classes
155 #####
156
157 #### https://parita-amin.github.io/survey.owl#CourseOffering
158 survey:CourseOffering rdf:type owl:Class ;
159                      rdfs:subClassOf survey:SurveyCollection ,
160                                     [ rdf:type owl:Restriction ;
161                                     owl:onProperty
162                                     survey:hasCollection ;
163                                     owl:someValuesFrom
164                                     survey:SurveyCollection
165                                     ] .
166
167 #### https://parita-amin.github.io/survey.owl#FeedbackSurvey
168 survey:FeedbackSurvey rdf:type owl:Class ;
169                      rdfs:subClassOf survey:Survey ;
170                      owl:disjointWith survey:FeedbackSurveyRespondent .
171
172 #### https://parita-amin.github.io/survey.owl#FeedbackSurveyRespondent
173 survey:FeedbackSurveyRespondent rdf:type owl:Class ;
174                      rdfs:subClassOf survey:SurveyRespondent .
175
176
177 ####
178 https://parita-amin.github.io/survey.owl#HeptingStudentFeedbackSurvey
179 survey:HeptingStudentFeedbackSurvey rdf:type owl:Class ;
180                      rdfs:subClassOf
181                      survey:FeedbackSurvey ;
182                      owl:disjointWith
183                      survey:HeptingStudentFeedbackSurveyRespondent .

```

```

181
182
183 ###
184     https://parita-amin.github.io/survey.owl#HeptingStudentFeedbackSurveyRespondent
185 survey:HeptingStudentFeedbackSurveyRespondent rdf:type owl:Class ;
186                                     rdfs:subClassOf
187     survey:FeedbackSurveyRespondent .
188
189 #### https://parita-amin.github.io/survey.owl#LSQ1
190 survey:LSQ1 rdf:type owl:Class ;
191             rdfs:subClassOf survey:LikertQuestion ,
192             [ rdf:type owl:Restriction ;
193               owl:onProperty survey:hasQuestionText ;
194               owl:hasValue "The instructor is
195               well-prepared for class"
196             ] .
197
198 #### https://parita-amin.github.io/survey.owl#LSQ1Ans
199 survey:LSQ1Ans rdf:type owl:Class ;
200               rdfs:subClassOf survey:LikertScaleAnswer .
201
202 #### https://parita-amin.github.io/survey.owl#LSQ2
203 survey:LSQ2 rdf:type owl:Class ;
204             rdfs:subClassOf survey:LikertQuestion ,
205             [ rdf:type owl:Restriction ;
206               owl:onProperty survey:hasQuestionText ;
207               owl:hasValue "The instructor clearly
208               communicates his expectations for student preparation and
209               participation"
210             ] .
211
212 #### https://parita-amin.github.io/survey.owl#LSQ2Ans
213 survey:LSQ2Ans rdf:type owl:Class ;
214               rdfs:subClassOf survey:LikertScaleAnswer .
215
216 #### https://parita-amin.github.io/survey.owl#LSQ3
217 survey:LSQ3 rdf:type owl:Class ;
218             rdfs:subClassOf survey:LikertQuestion ,
219             [ rdf:type owl:Restriction ;
220               owl:onProperty survey:hasQuestionText ;
221               owl:hasValue "The instructor uses class
222               time effectively"
223             ] .
224
225 #### https://parita-amin.github.io/survey.owl#LSQ3Ans
226 survey:LSQ3Ans rdf:type owl:Class ;

```

```

227         rdfs:subClassOf survey:LikertScaleAnswer .
228
229
230 #### https://parita-amin.github.io/survey.owl#LSQ4
231 survey:LSQ4 rdf:type owl:Class ;
232         rdfs:subClassOf survey:LikertQuestion ,
233         [ rdf:type owl:Restriction ;
234           owl:onProperty survey:hasQuestionText ;
235           owl:hasValue "The instructor has clear
236             expectations for assigned work"
237         ] .
238
239 #### https://parita-amin.github.io/survey.owl#LSQ4Ans
240 survey:LSQ4Ans rdf:type owl:Class ;
241         rdfs:subClassOf survey:LikertScaleAnswer .
242
243
244 #### https://parita-amin.github.io/survey.owl#LSQ5
245 survey:LSQ5 rdf:type owl:Class ;
246         rdfs:subClassOf survey:LikertQuestion ,
247         [ rdf:type owl:Restriction ;
248           owl:onProperty survey:hasQuestionText ;
249           owl:hasValue "The instructor encourages
250             student participation"
251         ] .
252
253 #### https://parita-amin.github.io/survey.owl#LSQ5Ans
254 survey:LSQ5Ans rdf:type owl:Class ;
255         rdfs:subClassOf survey:LikertScaleAnswer .
256
257
258 #### https://parita-amin.github.io/survey.owl#LSQ6
259 survey:LSQ6 rdf:type owl:Class ;
260         rdfs:subClassOf survey:LikertQuestion ,
261         [ rdf:type owl:Restriction ;
262           owl:onProperty survey:hasQuestionText ;
263           owl:hasValue "The instructor clearly
264             answers questions"
265         ] .
266
267 #### https://parita-amin.github.io/survey.owl#LSQ6Ans
268 survey:LSQ6Ans rdf:type owl:Class ;
269         rdfs:subClassOf survey:LikertScaleAnswer .
270
271
272 #### https://parita-amin.github.io/survey.owl#LSQ7
273 survey:LSQ7 rdf:type owl:Class ;
274         rdfs:subClassOf survey:LikertQuestion ,
275         [ rdf:type owl:Restriction ;

```

```

276                                owl:onProperty survey:hasQuestionText ;
277                                owl:hasValue "The instructor treats
    students with respect "
278                                ] .
279
280
281 #### https://parita-amin.github.io/survey.owl#LSQ7Ans
282 survey:LSQ7Ans rdf:type owl:Class ;
283                rdfs:subClassOf survey:LikertScaleAnswer .
284
285
286 #### https://parita-amin.github.io/survey.owl#LSQ8
287 survey:LSQ8 rdf:type owl:Class ;
288             rdfs:subClassOf survey:LikertQuestion ,
289                             [ rdf:type owl:Restriction ;
290                               owl:onProperty survey:hasQuestionText ;
291                               owl:hasValue "The instructor effectively
    directs and stimulates discussion "
292                             ] .
293
294
295 #### https://parita-amin.github.io/survey.owl#LSQ8Ans
296 survey:LSQ8Ans rdf:type owl:Class ;
297                rdfs:subClassOf survey:LikertScaleAnswer .
298
299
300 #### https://parita-amin.github.io/survey.owl#LSQ9
301 survey:LSQ9 rdf:type owl:Class ;
302             rdfs:subClassOf survey:LikertQuestion ,
303                             [ rdf:type owl:Restriction ;
304                               owl:onProperty survey:hasQuestionText ;
305                               owl:hasValue "The instructor effectively
    encourages students to ask questions and give answers "
306                             ] .
307
308
309 #### https://parita-amin.github.io/survey.owl#LSQ9Ans
310 survey:LSQ9Ans rdf:type owl:Class ;
311                rdfs:subClassOf survey:LikertScaleAnswer .
312
313
314 #### https://parita-amin.github.io/survey.owl#LikertQuestion
315 survey:LikertQuestion rdf:type owl:Class ;
316                       rdfs:subClassOf survey:LikertScaleQuestion ,
317                                       [ rdf:type owl:Restriction ;
318                                         owl:onProperty
    survey:hasLikertScale ;
319                                         owl:someValuesFrom
    survey:LikertScale
320                                       ] .
321
322

```



```

323 ### https://parita-amin.github.io/survey.owl#LikertScale
324 survey:LikertScale rdf:type owl:Class ;
325         owl:equivalentClass [ rdf:type owl:Class ;
326         owl:oneOf (
327         <https://parita-amin.github.io/survey.owl#1>
328         <https://parita-amin.github.io/survey.owl#2>
329         <https://parita-amin.github.io/survey.owl#3>
330         <https://parita-amin.github.io/survey.owl#4>
331         <https://parita-amin.github.io/survey.owl#5>
332         )
333         ] ;
334         rdfs:subClassOf survey:LikertScaleQuestion .
335
336 ### https://parita-amin.github.io/survey.owl#LikertScaleAnswer
337 survey:LikertScaleAnswer rdf:type owl:Class ;
338         rdfs:subClassOf survey:SurveyAnswer ,
339         [ rdf:type owl:Restriction ;
340         owl:onProperty
341         survey:hasLikertScale ;
342         owl:someValuesFrom
343         survey:LikertScale
344         ] ;
345         owl:disjointWith survey:OpenEndedAnswer .
346
347 ### https://parita-amin.github.io/survey.owl#LikertScaleQuestion
348 survey:LikertScaleQuestion rdf:type owl:Class ;
349         rdfs:subClassOf survey:SurveyQuestion ,
350         [ rdf:type owl:Restriction ;
351         owl:onProperty
352         survey:hasLikertScale ;
353         owl:qualifiedCardinality
354         "1"^^xsd:nonNegativeInteger ;
355         owl:onClass
356         survey:LikertScale
357         ] ;
358         owl:disjointWith survey:OpenEndedQuestion .
359
360 ### https://parita-amin.github.io/survey.owl#OEQ1
361 survey:OEQ1 rdf:type owl:Class ;
362         rdfs:subClassOf survey:OpenEndedQuestion ,
363         [ rdf:type owl:Restriction ;
364         owl:onProperty survey:hasQuestionText ;
365         owl:hasValue "What do you like best about
the course?"
366         ] .

```

```

364
365
366 ### https://parita-amin.github.io/survey.owl#OEQ1Ans
367 survey:OEQ1Ans rdf:type owl:Class ;
368         rdfs:subClassOf survey:OpenEndedAnswer ,
369         [ rdf:type owl:Restriction ;
370           owl:onProperty survey:hasOEQ1Ans ;
371           owl:someValuesFrom xsd:string
372         ] .
373
374
375 ### https://parita-amin.github.io/survey.owl#OEQ2
376 survey:OEQ2 rdf:type owl:Class ;
377         rdfs:subClassOf survey:OpenEndedQuestion ,
378         [ rdf:type owl:Restriction ;
379           owl:onProperty survey:hasQuestionText ;
380           owl:hasValue "What would you like to
381             change about the course?"
382         ] .
383
384 ### https://parita-amin.github.io/survey.owl#OEQ2Ans
385 survey:OEQ2Ans rdf:type owl:Class ;
386         rdfs:subClassOf survey:OpenEndedAnswer ,
387         [ rdf:type owl:Restriction ;
388           owl:onProperty survey:hasOEQ2Ans ;
389           owl:someValuesFrom xsd:string
390         ] .
391
392
393 ### https://parita-amin.github.io/survey.owl#OEQ3
394 survey:OEQ3 rdf:type owl:Class ;
395         rdfs:subClassOf survey:OpenEndedQuestion ,
396         [ rdf:type owl:Restriction ;
397           owl:onProperty survey:hasQuestionText ;
398           owl:hasValue "What do you think the
399             instructor's greatest strengths are?"
400         ] .
401
402 ### https://parita-amin.github.io/survey.owl#OEQ3Ans
403 survey:OEQ3Ans rdf:type owl:Class ;
404         rdfs:subClassOf survey:OpenEndedAnswer ,
405         [ rdf:type owl:Restriction ;
406           owl:onProperty survey:hasOEQ3Ans ;
407           owl:someValuesFrom xsd:string
408         ] .
409
410
411 ### https://parita-amin.github.io/survey.owl#OpenEndedAnswer
412 survey:OpenEndedAnswer rdf:type owl:Class ;
413         rdfs:subClassOf survey:SurveyAnswer .

```

```

414
415
416 #### https://parita-amin.github.io/survey.owl#OpenEndedQuestion
417 survey:OpenEndedQuestion rdf:type owl:Class ;
418         rdfs:subClassOf survey:SurveyQuestion .
419
420
421 #### https://parita-amin.github.io/survey.owl#Survey
422 survey:Survey rdf:type owl:Class ;
423         rdfs:subClassOf [ rdf:type owl:Restriction ;
424                             owl:onProperty survey:hasQuestion ;
425                             owl:someValuesFrom survey:SurveyQuestion
426                             ] .
427
428
429 #### https://parita-amin.github.io/survey.owl#SurveyAnswer
430 survey:SurveyAnswer rdf:type owl:Class ;
431         rdfs:subClassOf
432             survey:HeptingStudentFeedbackSurveyRespondent .
433
434 #### https://parita-amin.github.io/survey.owl#SurveyCollection
435 survey:SurveyCollection rdf:type owl:Class ;
436         rdfs:subClassOf [ rdf:type owl:Restriction ;
437                             owl:onProperty
438             survey:hasEligibleRespondents ;
439                             owl:someValuesFrom
440             survey:SurveyRespondent
441                             ] .
442
443 #### https://parita-amin.github.io/survey.owl#SurveyQuestion
444 survey:SurveyQuestion rdf:type owl:Class ;
445         rdfs:subClassOf
446             survey:HeptingStudentFeedbackSurvey ,
447             [ rdf:type owl:Restriction ;
448                 owl:onProperty
449                     survey:hasQuestionText ;
450                     owl:qualifiedCardinality
451                         "1"^^xsd:nonNegativeInteger ;
452                     owl:onDataRange xsd:string
453                     ] .
454
455 #### https://parita-amin.github.io/survey.owl#SurveyRespondent
456 survey:SurveyRespondent rdf:type owl:Class ;
457         rdfs:subClassOf [ rdf:type owl:Restriction ;
                             owl:onProperty
458             survey:hasCollection ;
459                             owl:someValuesFrom
460             survey:SurveyCollection
461                             ] .

```

```

458
459
460 #####
461 #      Individuals
462 #####
463
464 #### https://parita-amin.github.io/survey.owl#1
465 <https://parita-amin.github.io/survey.owl#1> rdf:type
      owl:NamedIndividual ,
466
      survey:LikertScale
467
468
469 #### https://parita-amin.github.io/survey.owl#2
470 <https://parita-amin.github.io/survey.owl#2> rdf:type
      owl:NamedIndividual ,
471
      survey:LikertScale
472
473
474 #### https://parita-amin.github.io/survey.owl#3
475 <https://parita-amin.github.io/survey.owl#3> rdf:type
      owl:NamedIndividual ,
476
      survey:LikertScale
477
478
479 #### https://parita-amin.github.io/survey.owl#4
480 <https://parita-amin.github.io/survey.owl#4> rdf:type
      owl:NamedIndividual ,
481
      survey:LikertScale
482
483
484 #### https://parita-amin.github.io/survey.owl#5
485 <https://parita-amin.github.io/survey.owl#5> rdf:type
      owl:NamedIndividual ,
486
      survey:LikertScale
487
488
489 #####
490 #      General axioms
491 #####
492
493 [ rdf:type owl:AllDisjointClasses ;
494   owl:members ( survey:LSQ1
495                  survey:LSQ2
496                  survey:LSQ3
497                  )
498 ] .
499

```

```

500
501 [ rdf:type owl:AllDisjointClasses ;
502   owl:members ( survey:LSQ1
503                   survey:LSQ2
504                   survey:LSQ3
505                   survey:LSQ4
506                   survey:LSQ5
507                   survey:LSQ6
508                   survey:LSQ7
509                   survey:LSQ8
510                   survey:LSQ9
511                 )
512 ] .
513
514
515 [ rdf:type owl:AllDisjointClasses ;
516   owl:members ( survey:LSQ1Ans
517                   survey:LSQ2Ans
518                   survey:LSQ3Ans
519                   survey:LSQ4Ans
520                   survey:LSQ5Ans
521                   survey:LSQ6Ans
522                   survey:LSQ7Ans
523                   survey:LSQ8Ans
524                   survey:LSQ9Ans
525                 )
526 ] .
527
528
529 [ rdf:type owl:AllDisjointClasses ;
530   owl:members ( survey:OEQ1
531                   survey:OEQ2
532                   survey:OEQ3
533                 )
534 ] .
535
536
537 [ rdf:type owl:AllDisjointClasses ;
538   owl:members ( survey:Survey
539                   survey:SurveyCollection
540                   survey:SurveyRespondent
541                 )
542 ] .
543
544
545 ### Generated by the OWL API (version 4.5.9.2019-02-01 T07:24:44Z)
      https://github.com/owlcs/owlapi

```

Appendix B

CSV to Linked Data: CS-280-202210.ttl

```
1 @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
2 @prefix owl: <http://www.w3.org/2002/07/owl#> .
3 @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
4 @prefix survey: <https://parita-amin.github.io/survey.owl#> .
5 @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
6 @prefix foaf: <http://xmlns.com/foaf/0.1/> .
7 @prefix : <https://parita-amin.github.io/survey-data/CS-280-202210.ttl#> .
8
9 :HeptingStudentFeedbackSurvey a survey:FeedbackSurvey;
10     survey:hasCourseOffering :CS-280-202210.
11
12 :CS-280-202210 a survey:CourseOffering;
13     survey:hasCollection :Midterm.
14
15 :Midterm a survey:SurveyCollection;
16     survey:hasEligibleRespondents "100"^^xsd:int;
17     survey:hasRespondent
18         :C1R1,
19         :C1R2,
20         :C1R3,
21         :C1R4,
22         :C1R5,
23         :C1R6,
24         :C1R7,
25         :C1R8,
26         :C1R9,
27         :C1R10,
28         :C1R11,
29         :C1R12,
30         :C1R13,
31         :C1R14,
32         :C1R15,
33         :C1R16,
```

```

34 :C1R17 ,
35 :C1R18 ,
36 :C1R19 ,
37 :C1R20 ,
38 :C1R21 ,
39 :C1R22 ,
40 :C1R23 ,
41 :C1R24 ,
42 :C1R25 ,
43 :C1R26 ,
44 :C1R27 ,
45 :C1R28 ,
46 :C1R29 ,
47 :C1R30 ,
48 :C1R31 ,
49 :C1R32 ,
50 :C1R33 ,
51 :C1R34 ,
52 :C1R35 ,
53 :C1R36 ,
54 :C1R37 ,
55 :C1R38 ,
56 :C1R39 ,
57 :C1R40 ,
58 :C1R41 ,
59 :C1R42 ,
60 :C1R43 ,
61 :C1R44 ,
62 :C1R45 ,
63 :C1R46 ,
64 :C1R47 ,
65 :C1R48 ,
66 :C1R49 ,
67 :C1R50 ,
68 :C1R51 ,
69 :C1R52 ,
70 :C1R53 ,
71 :C1R54 ,
72 :C1R55 ,
73 :C1R56 ,
74 :C1R57 ,
75 :C1R58 ,
76 :C1R59 .
77
78 :C1R1 a survey:SurveyRespondent ;
79 survey:hasLSQ6Ans "1"^^xsd:positiveInteger ;
80 survey:hasLSQ2Ans "1"^^xsd:positiveInteger ;
81 survey:hasLSQ8Ans "1"^^xsd:positiveInteger ;
82 survey:hasLSQ9Ans "1"^^xsd:positiveInteger ;
83 survey:hasLSQ5Ans "3"^^xsd:positiveInteger ;
84 survey:hasLSQ4Ans "1"^^xsd:positiveInteger ;
85 survey:hasLSQ1Ans "2"^^xsd:positiveInteger ;

```

```

86  survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
87  survey:hasLSQ3Ans "1"^^xsd:positiveInteger;
88  survey:hasOEQ2Ans "the professor , the grading scheme";
89  survey:hasOEQ1Ans "literally not a thing";
90  survey:hasOEQ3Ans "??".
91
92 :C1R2 a survey:SurveyRespondent;
93  survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
94  survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
95  survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
96  survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
97  survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
98  survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
99  survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
100 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
101 survey:hasLSQ3Ans "2"^^xsd:positiveInteger;
102 survey:hasOEQ2Ans "I feel like if we had some more information about
    debate topics we would be able to debate more effectively.";
103 survey:hasOEQ1Ans "I like how quick the teacher is to answer any
    question about a assignment or a project and clarify any detail.";
104 survey:hasOEQ3Ans "The teacher always answers questions about
    assignments and projects when ever he is asked about them so I always
    feel prepared for the next assignment." .
105
106 :C1R3 a survey:SurveyRespondent;
107  survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
108  survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
109  survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
110  survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
111  survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
112  survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
113  survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
114  survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
115  survey:hasLSQ3Ans "4"^^xsd:positiveInteger .
116
117 :C1R4 a survey:SurveyRespondent;
118  survey:hasLSQ6Ans "3"^^xsd:positiveInteger;
119  survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
120  survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
121  survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
122  survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
123  survey:hasLSQ4Ans "2"^^xsd:positiveInteger;
124  survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
125  survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
126  survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
127  survey:hasOEQ2Ans "I am confused on what the topics of this course
    even are. Every class it's students asking questions about the
    assignments and projects and then we have about 15 minutes at the end
    of every class to try and have a discussion.";
128  survey:hasOEQ3Ans "Having open communication with the students. No one
    is afraid to turn their microphones on and talk or share an opinion.
    It is a very welcoming environment." .

```



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129
130 :C1R5 a survey:SurveyRespondent;
131   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
132   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
133   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
134   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
135   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
136   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
137   survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
138   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
139   survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
140   survey:hasOEQ2Ans "This class is different from all the other classes
      I have taken. The discussions are open-ended and sometimes I feel I
      am not sure how well I am doing in this class or what I am learning.
      I would like to see more concrete implementation for achieving the
      desired learning outcomes. I do not what would be there on the final
      exam and what should I need to do to pass the final exam.";
141   survey:hasOEQ1Ans "The course effectively encourages me to ask
      questions and give answers. I like the class discussions on current
      topics in the world, helps me keep myself aware of what is happening
      in the world.";
142   survey:hasOEQ3Ans "Dr. Daryl Hepting is a great mentor with a lot of
      knowledge and experience teaching this course. He is always
      well-prepared for the class and communicates his expectations for
      students clearly. His teaching style encourages student
      participation. He is very respectful of his students and always
      willing to go above and beyond for helping his students. I would
      definitely take another class with Dr. Hepting." .
143
144 :C1R6 a survey:SurveyRespondent;
145   survey:hasLSQ6Ans "3"^^xsd:positiveInteger;
146   survey:hasLSQ2Ans "2"^^xsd:positiveInteger;
147   survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
148   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
149   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
150   survey:hasLSQ4Ans "2"^^xsd:positiveInteger;
151   survey:hasLSQ1Ans "2"^^xsd:positiveInteger;
152   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
153   survey:hasLSQ3Ans "1"^^xsd:positiveInteger;
154   survey:hasOEQ2Ans "The clarity on what is expected of us on
      assignments.";
155   survey:hasOEQ1Ans "The discussion section of each lecture.";
156   survey:hasOEQ3Ans "discussions" .
157
158 :C1R7 a survey:SurveyRespondent;
159   survey:hasLSQ6Ans "1"^^xsd:positiveInteger;
160   survey:hasLSQ2Ans "1"^^xsd:positiveInteger;
161   survey:hasLSQ8Ans "2"^^xsd:positiveInteger;
162   survey:hasLSQ9Ans "2"^^xsd:positiveInteger;
163   survey:hasLSQ5Ans "3"^^xsd:positiveInteger;
164   survey:hasLSQ4Ans "2"^^xsd:positiveInteger;
165   survey:hasLSQ1Ans "1"^^xsd:positiveInteger;

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166 survey:hasLSQ7Ans "3"^^xsd:positiveInteger;
167 survey:hasLSQ3Ans "1"^^xsd:positiveInteger .
168
169 :C1R8 a survey:SurveyRespondent;
170 survey:hasLSQ6Ans "3"^^xsd:positiveInteger;
171 survey:hasLSQ2Ans "3"^^xsd:positiveInteger;
172 survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
173 survey:hasLSQ9Ans "3"^^xsd:positiveInteger;
174 survey:hasLSQ5Ans "3"^^xsd:positiveInteger;
175 survey:hasLSQ4Ans "3"^^xsd:positiveInteger;
176 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
177 survey:hasLSQ7Ans "3"^^xsd:positiveInteger;
178 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
179 survey:hasOEQ2Ans "Nothing";
180 survey:hasOEQ1Ans "The content";
181 survey:hasOEQ3Ans "His communication skills" .
182
183 :C1R9 a survey:SurveyRespondent;
184 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
185 survey:hasLSQ2Ans "3"^^xsd:positiveInteger;
186 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
187 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
188 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
189 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
190 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
191 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
192 survey:hasLSQ3Ans "4"^^xsd:positiveInteger .
193
194 :C1R10 a survey:SurveyRespondent;
195 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
196 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
197 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
198 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
199 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
200 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
201 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
202 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
203 survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
204 survey:hasOEQ2Ans "NO";
205 survey:hasOEQ1Ans "The teacher's lessons are very informative, and
    then you need to reply every day so that people can discuss the
    content of the daily lessons.";
206 survey:hasOEQ3Ans "NO" .
207
208 :C1R11 a survey:SurveyRespondent;
209 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
210 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
211 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
212 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
213 survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
214 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
215 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;

```

```

216 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
217 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
218 survey:hasOEQ2Ans "Not much honestly , I don't see a need to change
    something that works.";
219 survey:hasOEQ1Ans "I enjoy how the instructor is very knowledgeable
    and always available when questions need to be answered.";
220 survey:hasOEQ3Ans "His knowledge." .
221
222 :C1R12 a survey:SurveyRespondent;
223 survey:hasLSQ6Ans "3"^^xsd:positiveInteger;
224 survey:hasLSQ2Ans "3"^^xsd:positiveInteger;
225 survey:hasLSQ8Ans "2"^^xsd:positiveInteger;
226 survey:hasLSQ9Ans "2"^^xsd:positiveInteger;
227 survey:hasLSQ5Ans "3"^^xsd:positiveInteger;
228 survey:hasLSQ4Ans "3"^^xsd:positiveInteger;
229 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
230 survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
231 survey:hasLSQ3Ans "1"^^xsd:positiveInteger;
232 survey:hasOEQ2Ans "I'd like more focus and discussion on the course
    material and more focus overall. Too much time is spent talking about
    assignment submissions and not enough about topics relating to the
    course material.Discussions more suited for office hours routinely
    dominate class time and obstruct relevant discussion.";
233 survey:hasOEQ1Ans "N/A";
234 survey:hasOEQ3Ans "N/A" .
235
236 :C1R13 a survey:SurveyRespondent;
237 survey:hasLSQ6Ans "3"^^xsd:positiveInteger;
238 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
239 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
240 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
241 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
242 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
243 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
244 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
245 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
246 survey:hasOEQ2Ans "Nothing I can think of";
247 survey:hasOEQ1Ans "The discussions";
248 survey:hasOEQ3Ans "Creative" .
249
250 :C1R14 a survey:SurveyRespondent;
251 survey:hasLSQ6Ans "2"^^xsd:positiveInteger;
252 survey:hasLSQ2Ans "3"^^xsd:positiveInteger;
253 survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
254 survey:hasLSQ9Ans "3"^^xsd:positiveInteger;
255 survey:hasLSQ5Ans "3"^^xsd:positiveInteger;
256 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
257 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
258 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
259 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
260 survey:hasOEQ2Ans "I would probably change how to teach things having
    more of a concrete definition of different topics.";

```

```

261 survey:hasOEQ1Ans "I like the say we are able to give about assignment
    due dates.";
262 survey:hasOEQ3Ans "Greatest strengths are being able to communicate
    with students about assignment topics as well as the freedom he is
    able to give students in his class." .
263
264 :C1R15 a survey:SurveyRespondent;
265 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
266 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
267 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
268 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
269 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
270 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
271 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
272 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
273 survey:hasLSQ3Ans "5"^^xsd:positiveInteger .
274
275 :C1R16 a survey:SurveyRespondent;
276 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
277 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
278 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
279 survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
280 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
281 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
282 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
283 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
284 survey:hasLSQ3Ans "4"^^xsd:positiveInteger .
285
286 :C1R17 a survey:SurveyRespondent;
287 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
288 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
289 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
290 survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
291 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
292 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
293 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
294 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
295 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
296 survey:hasOEQ2Ans "I don't think there are any changed requirements.";
297 survey:hasOEQ1Ans "This course is best to study with Dr Hepting as he
    includes most of the stuff practical based.";
298 survey:hasOEQ3Ans "Knowledge, Teaching skills." .
299
300 :C1R18 a survey:SurveyRespondent;
301 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
302 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
303 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
304 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
305 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
306 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
307 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
308 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;

```

```

309 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
310 survey:hasOEQ2Ans "The only thing I would like to change about this
    course would be to have the course content a little bit more
    organized and fast-paced and focused on the final exam since it is
    25% of our final grade and also explain what would the final exam
    entail.";
311 survey:hasOEQ1Ans "I really like how the professor encourages open
    student discussion on current world events and topics related to the
    course. The lively exchange of ideas during the class makes for the
    majority of the course content and it is an innovative way of
    learning and encouraging student participation.";
312 survey:hasOEQ3Ans "The instructor's greatest strengths are:— Encourage
    student participation and make every student feel welcomed to join
    any on-going discussion.— Engaging students by related current world
    events to our course content.— Patience and enthusiasm during the
    class.— Clarity about assignments" .
313
314 :C1R19 a survey:SurveyRespondent;
315 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
316 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
317 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
318 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
319 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
320 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
321 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
322 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
323 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
324 survey:hasOEQ2Ans "I wish that the expectations for assignments were
    explained more thoroughly further in advance rather than a week
    before";
325 survey:hasOEQ1Ans "I like the wide variety of topics discussed";
326 survey:hasOEQ3Ans "Interacting with students and encouraging
    discussion" .
327
328 :C1R20 a survey:SurveyRespondent;
329 survey:hasLSQ6Ans "3"^^xsd:positiveInteger;
330 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
331 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
332 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
333 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
334 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
335 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
336 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
337 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
338 survey:hasOEQ2Ans "To have a more structured lesson plan and teaching,
    we don't always have textbook readings and we don't do any
    note-taking in class, so I'm a little lost on what will be on the
    exam. we tend to go in circles a little bit.";
339 survey:hasOEQ1Ans "The class discussions and his understanding of the
    student's being confused";
340 survey:hasOEQ3Ans "relating content to the real world and making us
    see and compare it, not to made-up examples, but things we interact

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    with daily." .
341
342 :C1R21 a survey:SurveyRespondent;
343   survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
344   survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
345   survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
346   survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
347   survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
348   survey:hasLSQ4Ans "3"^^xsd:positiveInteger;
349   survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
350   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
351   survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
352   survey:hasOEQ2Ans "I would like to talk more about stuff we are told
    to watch or read before class in actual class time.";
353   survey:hasOEQ1Ans "I like the course material and that discussion is
    encouraged.";
354   survey:hasOEQ3Ans "Very kind and willing to answer questions." .
355
356 :C1R22 a survey:SurveyRespondent;
357   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
358   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
359   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
360   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
361   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
362   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
363   survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
364   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
365   survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
366   survey:hasOEQ1Ans "The best thing I like about this course is that I
    hear my peers' points of view and I can build my aspect upon their
    points of view." .
367
368 :C1R23 a survey:SurveyRespondent;
369   survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
370   survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
371   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
372   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
373   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
374   survey:hasLSQ4Ans "3"^^xsd:positiveInteger;
375   survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
376   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
377   survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
378   survey:hasOEQ2Ans "More clearly laid out assignment details and
    expectations; Sometimes feels like there's a lot of dead space in
    class where time could be used more effectively.";
379   survey:hasOEQ1Ans "The interesting discussions about online activity
    and information.";
380   survey:hasOEQ3Ans "The ability to foster and encourage interesting in
    class discussion." .
381
382 :C1R24 a survey:SurveyRespondent;
383   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;

```

```

384 survey:hasLSQ2Ans "2"^^xsd:positiveInteger;
385 survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
386 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
387 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
388 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
389 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
390 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
391 survey:hasLSQ3Ans "2"^^xsd:positiveInteger;
392 survey:hasOEQ2Ans "I would like to get rid of the response to meeting
    part. It's just frustrating that we have to do it every lecture";
393 survey:hasOEQ1Ans "The writing part, I am not good at it, this course
    is a chance for me to practice my writing skills.The topic we
    discussed is very interesting tho.";
394 survey:hasOEQ3Ans "His preparation skills. I believe that Dr Hepting
    knew his weakness, so he created so many tools, so many notes that
    the students can use. And for a \"geek\" like me, they are amazing!" .
395
396 :C1R25 a survey:SurveyRespondent;
397 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
398 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
399 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
400 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
401 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
402 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
403 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
404 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
405 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
406 survey:hasOEQ2Ans "Nothing so far.";
407 survey:hasOEQ1Ans "The professor is very considerate of the students.
    He always encourages lively discussions.";
408 survey:hasOEQ3Ans "Instructor is very considerate of students. Does
    not give work during holidays or immediately after. Gives students
    time and second chances." .
409
410 :C1R26 a survey:SurveyRespondent;
411 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
412 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
413 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
414 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
415 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
416 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
417 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
418 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
419 survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
420 survey:hasOEQ2Ans "Nothing from my point of view.";
421 survey:hasOEQ1Ans "Professor Hepting I guess, he is well prepared and
    has answers to our questions promptly.";
422 survey:hasOEQ3Ans "Not going to a different topic unless things are
    clear for all the students." .
423
424 :C1R27 a survey:SurveyRespondent;
425 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;

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426 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
427 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
428 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
429 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
430 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
431 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
432 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
433 survey:hasLSQ3Ans "5"^^xsd:positiveInteger .
434
435 :C1R28 a survey:SurveyRespondent;
436 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
437 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
438 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
439 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
440 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
441 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
442 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
443 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
444 survey:hasLSQ3Ans "3"^^xsd:positiveInteger .
445
446 :C1R29 a survey:SurveyRespondent;
447 survey:hasLSQ6Ans "3"^^xsd:positiveInteger;
448 survey:hasLSQ2Ans "3"^^xsd:positiveInteger;
449 survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
450 survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
451 survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
452 survey:hasLSQ4Ans "3"^^xsd:positiveInteger;
453 survey:hasLSQ1Ans "2"^^xsd:positiveInteger;
454 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
455 survey:hasLSQ3Ans "2"^^xsd:positiveInteger;
456 survey:hasOEQ2Ans "i wish there was better organization and better use
    of class time.";
457 survey:hasOEQ1Ans "everything seems relatively easy.";
458 survey:hasOEQ3Ans "he has good topics to talk about." .
459
460 :C1R30 a survey:SurveyRespondent;
461 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
462 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
463 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
464 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
465 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
466 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
467 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
468 survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
469 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
470 survey:hasOEQ2Ans "Better ways of communication during class for all
    our classmates so breakout rooms go more smoothly.";
471 survey:hasOEQ1Ans "The fact that most of the topics are relevant to
    today's world and not outdated.";
472 survey:hasOEQ3Ans "His compassion towards the students and
    understanding our own struggles as well." .
473

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474 :C1R31 a survey:SurveyRespondent;
475   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
476   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
477   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
478   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
479   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
480   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
481   survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
482   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
483   survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
484   survey:hasOEQ2Ans "Nothing ! It's all good and perfect.";
485   survey:hasOEQ1Ans "It's interesting and non complicated course which
      is based on our daily lives. The main thing is that it does not
      require mental attention and it's a free learning and fun course.";
486   survey:hasOEQ3Ans "He is soft spoken, always keeps patience, answers
      questions and explains things thoroughly." .
487
488 :C1R32 a survey:SurveyRespondent;
489   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
490   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
491   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
492   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
493   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
494   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
495   survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
496   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
497   survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
498   survey:hasOEQ2Ans "Nothing in particular.";
499   survey:hasOEQ1Ans "The topics which are covered in this course.";
500   survey:hasOEQ3Ans "The greatest strength of the instructor of this
      course is that he ready to help even if you ask question in middle of
      discussion he is ready to answer. Also to make students more
      interactive he arranges class discussion as well which also with the
      participation." .
501
502 :C1R33 a survey:SurveyRespondent;
503   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
504   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
505   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
506   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
507   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
508   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
509   survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
510   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
511   survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
512   survey:hasOEQ2Ans "nothing";
513   survey:hasOEQ1Ans "The best part is how the instructor teaches and
      explains the stuff";
514   survey:hasOEQ3Ans "Explaining the course" .
515
516 :C1R34 a survey:SurveyRespondent;
517   survey:hasLSQ6Ans "4"^^xsd:positiveInteger;

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518 survey:hasLSQ2Ans "3"^^xsd:positiveInteger;
519 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
520 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
521 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
522 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
523 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
524 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
525 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
526 survey:hasOEQ2Ans "Lecture's are little boring and need some
    entertainment. He is a great teacher but students need some
    motivation thats is what is missing and a big smile is missing of
    professor";
527 survey:hasOEQ1Ans "The thing i like most about this course is it very
    based on research and that is what is making me research more about
    the topics discussed. Since then i have started to research more and
    more about the class";
528 survey:hasOEQ3Ans "He is really good with the topics and knows
    everything he teaches he helps us with some research strategies thats
    what is the greatest strength." .
529
530 :C1R35 a survey:SurveyRespondent;
531 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
532 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
533 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
534 survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
535 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
536 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
537 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
538 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
539 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
540 survey:hasOEQ2Ans "I'd change how the course is presented. Whether or
    not it is a teaching style preference, I tend to prefer classes that
    have a bit more structure to them, and less group activities as I
    prefer to rely solely on myself for my grade.";
541 survey:hasOEQ1Ans "I like the idea of the course the best. The idea of
    weighing risks vs rewards of giving companies your information.";
542 survey:hasOEQ3Ans "I think the instructor's greatest strength is his
    personal interest in the class. He seems to care deeply about the
    subject, enough to let it bleed into the class work, lectures, etc." .
543
544 :C1R36 a survey:SurveyRespondent;
545 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
546 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
547 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
548 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
549 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
550 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
551 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
552 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
553 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
554 survey:hasOEQ2Ans "Less group work, especially for a class delivered
    remotely." ;

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555 survey:hasOEQ1Ans "The topics discussed and the way assignments are
    delivered.";
556 survey:hasOEQ3Ans "The instructor is a very smart individual who is
    very passionate about the topics." .
557
558 :C1R37 a survey:SurveyRespondent;
559 survey:hasLSQ6Ans "2"^^xsd:positiveInteger;
560 survey:hasLSQ2Ans "2"^^xsd:positiveInteger;
561 survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
562 survey:hasLSQ9Ans "3"^^xsd:positiveInteger;
563 survey:hasLSQ5Ans "3"^^xsd:positiveInteger;
564 survey:hasLSQ4Ans "2"^^xsd:positiveInteger;
565 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
566 survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
567 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
568 survey:hasOEQ2Ans "Nothing";
569 survey:hasOEQ1Ans "My classmates and I continuously communicate with
    each other to help understand what's happening, ask questions about
    homework, and such. The topics can be quite interesting and with each
    discussion, I learn something new.";
570 survey:hasOEQ3Ans "He is knowledgable and well updated with tech
    news." .
571
572 :C1R38 a survey:SurveyRespondent;
573 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
574 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
575 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
576 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
577 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
578 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
579 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
580 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
581 survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
582 survey:hasOEQ2Ans "Honestly, getting up early is hard probably would
    like to change the class timing by a hour maybe";
583 survey:hasOEQ1Ans "The fact that our professor encourages more class
    participation by effective methods like group projects and always
    relate topics with current affairs is the best part of this course to
    me.";
584 survey:hasOEQ3Ans "Our instructors greatest strength his ability to
    relate the topic with current affairs." .
585
586 :C1R39 a survey:SurveyRespondent;
587 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
588 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
589 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
590 survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
591 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
592 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
593 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
594 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
595 survey:hasLSQ3Ans "2"^^xsd:positiveInteger;

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596 survey:hasOEQ2Ans "I would like to see more content in class. I feel
    we don't really get anything done in class till the end.";
597 survey:hasOEQ1Ans "I really like the content of this course. It is
    very interesting and changes my perspective of technology";
598 survey:hasOEQ3Ans "The instructor's greatest strengths are that he is
    very knowledgeable and genuinely cares for his students." .
599
600 :C1R40 a survey:SurveyRespondent;
601 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
602 survey:hasLSQ2Ans "1"^^xsd:positiveInteger;
603 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
604 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
605 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
606 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
607 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
608 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
609 survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
610 survey:hasOEQ2Ans "I dont like how we are supposed to work in
    different groups for each assessment which makes it confusing and
    unfair for the student with no connections or friends at all.";
611 survey:hasOEQ1Ans "Professor give equal opportunity to every student
    in the class and explain every single detail of our works we need to
    submit. Even present some examples of our submission to make it even
    more clear.";
612 survey:hasOEQ3Ans "Professor is a good listener and believes in team
    work." .
613
614 :C1R41 a survey:SurveyRespondent;
615 survey:hasLSQ6Ans "2"^^xsd:positiveInteger;
616 survey:hasLSQ2Ans "3"^^xsd:positiveInteger;
617 survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
618 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
619 survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
620 survey:hasLSQ4Ans "2"^^xsd:positiveInteger;
621 survey:hasLSQ1Ans "2"^^xsd:positiveInteger;
622 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
623 survey:hasLSQ3Ans "2"^^xsd:positiveInteger;
624 survey:hasOEQ2Ans "To be honest, I didn't find anything best yet,
    because we didn't get proper knowledge in this class as we have given
    the name of a textbook but in class, we didn't get any information
    regarding our course. Every day I have found that Professor is busy
    changing the due dates and editing on URCourses. One of my friends
    suggested CS280 and also told me that I will gain more knowledge as
    well as I will really enjoy this course. But, somehow I'm feeling
    completely opposite. I'm completely disappointed with this course.
    Hope I'll get at least some knowledge of this course and suggest
    others for this course.";
625 survey:hasOEQ1Ans "As this course is running remotely as well as
    in-person which is the only one best thing I have found. Also, The
    way the professor treats us with very polite behavior.";
626 survey:hasOEQ3Ans "I have found that the professor is more
    knowledgeable but his teaching style does not fit particularly in my

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mind." .
627
628 :C1R42 a survey:SurveyRespondent;
629   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
630   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
631   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
632   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
633   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
634   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
635   survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
636   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
637   survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
638   survey:hasOEQ2Ans "I would like to change the group studies portion to
        individual, so we don't have to be dependent on others.";
639   survey:hasOEQ1Ans "The delivery of the course is excellent";
640   survey:hasOEQ3Ans "The explanation skills and most importantly the
        period of giving time to students." .
641
642 :C1R43 a survey:SurveyRespondent;
643   survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
644   survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
645   survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
646   survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
647   survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
648   survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
649   survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
650   survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
651   survey:hasLSQ3Ans "4"^^xsd:positiveInteger .
652
653 :C1R44 a survey:SurveyRespondent;
654   survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
655   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
656   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
657   survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
658   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
659   survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
660   survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
661   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
662   survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
663   survey:hasOEQ2Ans "Nothing.";
664   survey:hasOEQ1Ans "Flexible assignment due dates. Different projects
        with a different perspective can make a lot of difference in group
        projects.";
665   survey:hasOEQ3Ans "The way he conducts the class and how he
        organizes his website like detailed assignment requirements and the
        syllabus also how and when the assignment and exams are due." .
666
667 :C1R45 a survey:SurveyRespondent;
668   survey:hasLSQ6Ans "3"^^xsd:positiveInteger;
669   survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
670   survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
671   survey:hasLSQ9Ans "4"^^xsd:positiveInteger;

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672 survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
673 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
674 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
675 survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
676 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
677 survey:hasOEQ2Ans "I would like to change how much down time there is
    in class, I think we spend too much time in between topics or
    discussions, and if we didn't do that I think everyone would be able
    to learn a lot more.";
678 survey:hasOEQ1Ans "I like that we can tie in class material with
    current events, because I think talking about relevant news is more
    interesting than just reading a textbook.";
679 survey:hasOEQ3Ans "I think that the professor's greatest strengths are
    his knowledge on the subject matter being taught, whatever we talk
    about he for the most part has a good idea on the concepts, but he
    can sometime struggle to explain it better for other students who
    didn't understand." .
680
681 :C1R46 a survey:SurveyRespondent;
682 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
683 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
684 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
685 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
686 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
687 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
688 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
689 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
690 survey:hasLSQ3Ans "1"^^xsd:positiveInteger;
691 survey:hasOEQ2Ans "While I like the idea of learning about risks and
    rewards in computer science it seems like we never actually learn
    anything in class. Instead we just go over what's expected on
    assignments and answer and re-answer the same questions. I wish we
    could actually learn something new for a change.";
692 survey:hasOEQ1Ans "I like how Dr. Hepting encourages conversation in
    class and encourages student participation.";
693 survey:hasOEQ3Ans "Listening to students and accommodating are needs."
    .
694
695 :C1R47 a survey:SurveyRespondent;
696 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
697 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
698 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
699 survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
700 survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
701 survey:hasLSQ4Ans "3"^^xsd:positiveInteger;
702 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
703 survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
704 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
705 survey:hasOEQ2Ans "I wish it was more constructed and had notes or
    readings for us to do";
706 survey:hasOEQ1Ans "The in class discussions are interesting" .
707

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708 :C1R48 a survey:SurveyRespondent;
709   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
710   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
711   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
712   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
713   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
714   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
715   survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
716   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
717   survey:hasLSQ3Ans "5"^^xsd:positiveInteger .
718
719 :C1R49 a survey:SurveyRespondent;
720   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
721   survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
722   survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
723   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
724   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
725   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
726   survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
727   survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
728   survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
729   survey:hasOEQ2Ans "big writing assignments.";
730   survey:hasOEQ3Ans "the ability to teach." .
731
732 :C1R50 a survey:SurveyRespondent;
733   survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
734   survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
735   survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
736   survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
737   survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
738   survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
739   survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
740   survey:hasLSQ7Ans "4"^^xsd:positiveInteger;
741   survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
742   survey:hasOEQ2Ans "More assignments.";
743   survey:hasOEQ1Ans "The topics are really interesting.";
744   survey:hasOEQ3Ans "Content knowledge and experience." .
745
746 :C1R51 a survey:SurveyRespondent;
747   survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
748   survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
749   survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
750   survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
751   survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
752   survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
753   survey:hasLSQ1Ans "4"^^xsd:positiveInteger;
754   survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
755   survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
756   survey:hasOEQ2Ans "I'd like to spend more time exploring topics and
      learning new ideas than just administration stuff. I think it's only
      like this because of online learning and Covid, though, so I
      understand if there's a lot of difficulties we have to work through.

```

I just wish it wasn't so often. It'll probably get better once everything is in person.";

757 survey:hasOEQ1Ans "The discussion on various topics; gaining new insight and ideas from my peers; Very low-intensity; Lots of critical thinking and research; I also really like the post-lecture feedback from each student regarding that day's topic. It's interesting to see other people's posts because I can learn new ideas from them.";

758 survey:hasOEQ3Ans "Really engages class discussion. It's nice because you get to hear other views Provides some really good resources (all of which are free!). The Ted talks, articles, and book are pretty relevant and insightful. I appreciate that everything is gathered together in the website because I might miss something. I also appreciate the class notes summary that students post. I can look at them in case I miss a class." .

759

760 :C1R52 a survey:SurveyRespondent;

761 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;

762 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;

763 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;

764 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;

765 survey:hasLSQ5Ans "4"^^xsd:positiveInteger;

766 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;

767 survey:hasLSQ1Ans "4"^^xsd:positiveInteger;

768 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;

769 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;

770 survey:hasOEQ2Ans "I think there is nothing that needs to be changed except giving lot of assignments..";

771 survey:hasOEQ1Ans "Taking this course I got to know many things like how information is important and what is privacy and so on... So it is an interesting class I guess...";

772 survey:hasOEQ3Ans "I think Dr Daryl Hepting has good nature and he is friendly with everyone and so I think this is his greatest strength..." .

773

774 :C1R53 a survey:SurveyRespondent;

775 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;

776 survey:hasLSQ2Ans "2"^^xsd:positiveInteger;

777 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;

778 survey:hasLSQ9Ans "3"^^xsd:positiveInteger;

779 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;

780 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;

781 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;

782 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;

783 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;

784 survey:hasOEQ2Ans "I would make the expectations and how to approach and succeed in this course clear at the beginning of the course.";

785 survey:hasOEQ1Ans "I like how there's no coding in this computer science course.";

786 survey:hasOEQ3Ans "The instructor greatest strength is his ability to start a discussion among students" .

787

788 :C1R54 a survey:SurveyRespondent;


```

789 survey:hasLSQ6Ans "4"^^xsd:positiveInteger;
790 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
791 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
792 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
793 survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
794 survey:hasLSQ4Ans "4"^^xsd:positiveInteger;
795 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
796 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
797 survey:hasLSQ3Ans "3"^^xsd:positiveInteger;
798 survey:hasOEQ2Ans "If we could be more on topic in most lectures ,
    that'd be great. We spend so much time talking about assignments that
    the actual lectures and discussions feel few and far between compared
    to the course admin days.";
799 survey:hasOEQ1Ans "I enjoy the discussions the most honestly , hearing
    everyone's opinions on web safety topics (tracking , articles and
    truth) is probably one of my favorite things about this course.";
800 survey:hasOEQ3Ans "The website used for documentation about meeting
    documentation (transcripts , responses) and assignments is very nice ,
    and I'm glad effort has gone into making it a good website. While
    stuttering a lot makes it a bit hard to understand what he's saying ,
    it's not like it makes what he's saying any less interesting. I
    usually end up thinking about the discussion topics or the lecture
    topic far after the class because they're that interesting." .
801
802 :C1R55 a survey:SurveyRespondent;
803 survey:hasOEQ2Ans "more clarity";
804 survey:hasOEQ1Ans "variety of topics" .
805
806 :C1R56 a survey:SurveyRespondent;
807 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;
808 survey:hasLSQ2Ans "5"^^xsd:positiveInteger;
809 survey:hasLSQ8Ans "5"^^xsd:positiveInteger;
810 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
811 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
812 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
813 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
814 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
815 survey:hasLSQ3Ans "4"^^xsd:positiveInteger;
816 survey:hasOEQ2Ans "This course is good. The only thing is sometimes Dr
    Daryl don't reply to emails.";
817 survey:hasOEQ1Ans "The freedom to speak what you think and believe in.
    The freedom and encouragement of expressing your thoughts and
    opinions about various topics. I like the professor. Dr Daryl always
    keeps his calm. I have never seen him angry; I like his patience. He
    listens to the students , for example when we asked for extending the
    deadlines for blog entry assignment , he listened to us and extended
    it.";
818 survey:hasOEQ3Ans "His calm , patience , knowledge , enthusiasm about
    participation." .
819
820 :C1R57 a survey:SurveyRespondent;
821 survey:hasLSQ6Ans "5"^^xsd:positiveInteger;

```

```

822 survey:hasLSQ2Ans "4"^^xsd:positiveInteger;
823 survey:hasLSQ8Ans "4"^^xsd:positiveInteger;
824 survey:hasLSQ9Ans "5"^^xsd:positiveInteger;
825 survey:hasLSQ5Ans "5"^^xsd:positiveInteger;
826 survey:hasLSQ4Ans "5"^^xsd:positiveInteger;
827 survey:hasLSQ1Ans "5"^^xsd:positiveInteger;
828 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
829 survey:hasLSQ3Ans "5"^^xsd:positiveInteger;
830 survey:hasOEQ2Ans "If I want to change something in this course would
    be the appropriate syllabus for the final exam. And the other thing
    would be that I would be glad if this course would be in person
    because, for me, offline learning is more efficient than online
    learning. :)";
831 survey:hasOEQ1Ans "We get to know about a lot of things from this
    course. That, in this information society there are both risks and
    rewards. It depends on us how we use this. Using the internet and
    other software with full knowledge is quite important to earn maximum
    rewards. But, if we don't use it securely then it can be harmful to
    us and can disturb our private life. The professor is awesome as he
    enforces more participation from students and listens to their points
    and views carefully and corrects them where ever possible. Thanks for
    this wonderful course and a great professor. :)";
832 survey:hasOEQ3Ans "The instructor's greater strength is that he does
    not only speak and teach, he also expects students to come up and
    discuss with him their views and he listens to us very carefully and
    always comes up with a better idea to make us understand more
    properly. Other than that he is a great tutor and I love attending
    his class. Thank you, professor. :)".
833
834 :C1R58 a survey:SurveyRespondent;
835 survey:hasLSQ6Ans "2"^^xsd:positiveInteger;
836 survey:hasLSQ2Ans "3"^^xsd:positiveInteger;
837 survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
838 survey:hasLSQ9Ans "4"^^xsd:positiveInteger;
839 survey:hasLSQ5Ans "3"^^xsd:positiveInteger;
840 survey:hasLSQ4Ans "3"^^xsd:positiveInteger;
841 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
842 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
843 survey:hasLSQ3Ans "3"^^xsd:positiveInteger.
844
845 :C1R59 a survey:SurveyRespondent;
846 survey:hasLSQ6Ans "2"^^xsd:positiveInteger;
847 survey:hasLSQ2Ans "1"^^xsd:positiveInteger;
848 survey:hasLSQ8Ans "3"^^xsd:positiveInteger;
849 survey:hasLSQ9Ans "3"^^xsd:positiveInteger;
850 survey:hasLSQ5Ans "4"^^xsd:positiveInteger;
851 survey:hasLSQ4Ans "3"^^xsd:positiveInteger;
852 survey:hasLSQ1Ans "3"^^xsd:positiveInteger;
853 survey:hasLSQ7Ans "5"^^xsd:positiveInteger;
854 survey:hasLSQ3Ans "1"^^xsd:positiveInteger;
855 survey:hasOEQ2Ans "1. I'd like a roadmap to final exam prep – With the
    final coming up it is important to have a clear understanding of how

```

we should be preparing for it. Would it be fair to say that the majority of the test will be on the textbook? Because as of right now I would be able to answer none of the practice exam questions because we haven't covered the relevant events — which has me concerned that the final — which I have to pass to pass the class — may be the same since practice exams are supposed to help you practice for the final. 2. Pacing/speaking skills/preparation could improve — I was surprised when I joined this class. Your speech was very slow and you seemed to be struggling to communicate. Since that day, every day you have gotten slightly better at speaking. I just wanted to acknowledge that this is a thing and many people were concerned about it. Several people I know dropped your class because of it both this year and previous years. I'm not sure what was causing this, but I thought you ought to know. I will also add that today your speech seemed totally normal and you were prepared and had good pacing.";

856 survey:hasOEQ1Ans "It opened my eyes to the act of analyzing algorithms to ensure they are ethical.";

857 survey:hasOEQ3Ans "You listen to people's feedback during class and seem to actually care about people. That goes a long way." .

Appendix C

SHACL Shapes for Data Validation

This file (<https://parita-amin.github.io/SurveyShapes.txt>) contains the SHACL description for data validation according to the survey.owl vocabulary.

```
1 @prefix rdf:    <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
2 @prefix sh:     <http://www.w3.org/ns/shacl#> .
3 @prefix xsd:    <http://www.w3.org/2001/XMLSchema#> .
4 @prefix rdfs:   <http://www.w3.org/2000/01/rdf-schema#> .
5 @prefix survey: <https://parita-amin.github.io/survey.owl#> .
6 @prefix owl:  <http://www.w3.org/2002/07/owl#> .
7
8 survey:RespondentShape
9   a sh:NodeShape ;
10    sh:targetClass survey:SurveyRespondent ; # Applies to all Respondents
11
12    sh:property [
13      sh:path survey:hasLSQ6Ans ;          # constrains the values of
survey:hasLSQ6Ans
14      sh:datatype xsd:positiveInteger ;
15      sh:minInclusive 1 ;
16      sh:maxInclusive 5 ;
17      sh:severity sh:Warning ;
18    ] ;
19
20    sh:property [
21      sh:path survey:hasLSQ2Ans ;          # constrains the values of
survey:hasLSQ2Ans
22      sh:datatype xsd:positiveInteger ;
23      sh:minInclusive 1 ;
24      sh:maxInclusive 5 ;
25      sh:severity sh:Warning ;
26    ] ;
27
28    sh:property [
29      sh:path survey:hasLSQ8Ans ;          # constrains the values of
survey:hasLSQ8Ans
30      sh:datatype xsd:positiveInteger ;
31      sh:minInclusive 1 ;
```

```

32         sh:maxInclusive 5 ;
33         sh:severity sh:Warning ;
34     ] ;
35
36     sh:property [
37         sh:path survey:hasLSQ9Ans ;           # constrains the values of
survey:hasLSQ9Ans
38         sh:datatype xsd:positiveInteger ;
39         sh:minInclusive 1 ;
40         sh:maxInclusive 5 ;
41         sh:severity sh:Warning ;
42     ] ;
43
44     sh:property [
45         sh:path survey:hasLSQ5Ans ;           # constrains the values of
survey:hasLSQ5Ans
46         sh:datatype xsd:positiveInteger ;
47         sh:minInclusive 1 ;
48         sh:maxInclusive 5 ;
49         sh:severity sh:Warning ;
50     ] ;
51
52     sh:property [
53         sh:path survey:hasLSQ4Ans ;           # constrains the values of
survey:hasLSQ4Ans
54         sh:datatype xsd:positiveInteger ;
55         sh:minInclusive 1 ;
56         sh:maxInclusive 5 ;
57         sh:severity sh:Warning ;
58     ] ;
59
60     sh:property [
61         sh:path survey:hasLSQ1Ans ;           # constrains the values of
survey:hasLSQ1Ans
62         sh:datatype xsd:positiveInteger ;
63         sh:minInclusive 1 ;
64         sh:maxInclusive 5 ;
65         sh:severity sh:Warning ;
66     ] ;
67
68     sh:property [
69         sh:path survey:hasLSQ7Ans ;           # constrains the values of
survey:hasLSQ7Ans
70         sh:datatype xsd:positiveInteger ;
71         sh:minInclusive 1 ;
72         sh:maxInclusive 5 ;
73         sh:severity sh:Warning ;
74     ] ;
75
76     sh:property [
77         sh:path survey:hasLSQ3Ans ;           # constrains the values of
survey:hasLSQ3Ans

```

```
78         sh:datatype xsd:positiveInteger ;
79         sh:minInclusive 1 ;
80         sh:maxInclusive 5 ;
81         sh:severity sh:Warning ;
82     ] ;
83
84     sh:closed false ;
85     sh:ignoredProperties ( rdf:type owl:topDataProperty
86                           owl:topObjectProperty ) ;
86     .
87 ##### add SHACL vocabulary #####
```